Group 13: Boron Family ($ns^2\ np^1$) — Periodic Trends & Anomalies

Group 13 contains B (non-metal), Al (metal/amphoteric), Ga, In, Tl (post-transition metals). The chemistry is dominated by: small size of B, inert pair effect descending the group, and poor d/f electron shielding causing atomic radius and ionization enthalpy anomalies.

1. Atomic Radii & Ionization Enthalpy Anomalies

Element	IE-1 (kJ/mol)	Atomic Radius (pm)	Key Electronic Reason
B [He] $2s^2\ 2p^1$	801	85	Extremely small size, covalent network, high charge density
Al [Ne] $3s^2\ 3p^1$	577	143	Normal expansion with full 8-electron noble core
Ga [Ar] $3d^{10}\ 4s^2\ 4p^1$	579	135	r(Ga) < r(Al)! Poor shielding by 3d^10 → Z_eff increases
In [Kr] $4d^{10}\ 5s^2\ 5p^1$	558	167	Normal size increase after 4d shell
Tl [Xe] $4f^{14}\ 5d^{10}\ 6s^2\ 6p^1$	589	170	IE(Tl) > IE(In)! Lanthanoid contraction (poor 4f shielding)

Table 13.1 — G13 physical property anomalies (NCERT & JD Lee).

KEY RESULT
Atomic Radius Order: **B Tl > Ga > Al > In** (Tl has 2nd highest IE due to 4f contraction).

2. Inert Pair Effect & Oxidation States (+3 vs +1)

Descending the group, the 6s^2 pair of valence electrons becomes increasingly reluctant to participate in bonding due to high effective nuclear charge following 4f and 5d subshells.

- **+3 Stability Order:** B(+3) > Al(+3) > Ga(+3) > In(+3) > Tl(+3). Tl^{3+} is a powerful **oxidising agent** ($Tl^{3+} + 2e^- \rightarrow Tl^+$, $E^\circ = +1.26$ V).
- **+1 Stability Order:** B(+1) < Al(+1) < Ga(+1) < In(+1) < **Tl(+1)**. TlCl, TlOH are ionic and stable like alkali metals (TlOH is a strong base).
- **Al(+1) and Ga(+1):** Disproportionate in water: $3\ Ga^+ \rightarrow 2\ Ga + Ga^{3+}$.

TRAP
TRAP: TlI3 is NOT Thallium(III) iodide! It is an ionic compound **$Tl^+ (I_3^-)$** containing the linear triiodide anion. Tl^{3+} readily oxidises I^- to I_2!

3. Extraction & Reactivity of Boron

Boron is extracted from **Borax ($Na_2B_4O_7 \cdot 10H_2O$)** or **Colemanite ($Ca_2B_6O_{11} \cdot 5H_2O$)**: 1. Boil ore with dil. H_2SO_4 → precipitate orthoboric acid H_3BO_3. 2. Ignite $H_3BO_3 \rightarrow B_2O_3$. Then reduce: $B_2O_3 + 3$ Mg → 2 B (amorphous) + 3 MgO. 3. Ultra-pure crystalline B via van Arkel: 2 BBr3 + 3 H_2 (Ta filament, 1200°C) → 2 B + 6 HBr.

Diborane (B_2H_6) — Synthesis, 3c-2e Bonding, Cleavage & Reactions

Boron has only 3 valence electrons, so monomeric BH3 has only 6 valence electrons (Lewis acid). It dimerises to B_2H_6 via two electron-deficient bridging bonds (multicentre 3c-2e bonds).

1. Structure & Multi-Centre Bonding of B2H6

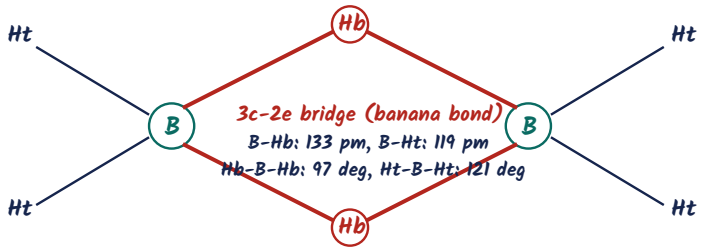


Fig. 13.1 — Diborane geometry with terminal (Ht) and bridging (Hb) hydrogens.

- **Terminal B-Ht:** 4 normal covalent 2c-2e bonds, coplanar in the xy-plane (B-Ht = 119 pm, H-B-H = 121°).
- **Bridging B-Hb-B:** 2 three-centre two-electron (3c-2e) 'banana' bonds, lying above and below the xy-plane (B-Hb = 133 pm, B-Hb-B = 97°).
- **Hybridization of B:** Each Boron atom is **sp^3 hybridized** (4 electron domains). Total valence electrons = 2×3 + 6×1 = 12 electrons (6 pairs for 8 bond vectors).

2. Cleavage of B2H6: Symmetrical vs Unsymmetrical

The mode of Lewis base cleavage depends directly on the base's steric size and hardness:

Type of Cleavage	Reagents (Lewis Bases)	Product & Reaction Equation
Symmetrical Cleavage	Bulky/Soft bases: NMe3, THF, CO, SMe2, PMe3	$B_2H_6 + 2\ NMe3 \rightarrow 2\ Me_3N{:}BH3$ (splits into two BH3·L)
Unsymmetrical Cleavage	Small/Hard bases: NH3 (low T), MeNH2, Me_2NH	$B_2H_6 + 2\ NH3 \rightarrow [H_2B(NH_3)_2]^+\ [BH4]^-$ (ionic salt)

Table 13.2 — Cleavage mechanisms of Diborane (NCERT & JD Lee core).

KEY RESULT
Industrial Prep: $2\ BF3 + 6\ NaH \rightarrow B_2H_6 + 6\ NaF$ (at 450 K in polyether). Laboratory Prep: $4\ BF3{\cdot}OEt2 + 3\ LiAlH4 \rightarrow 2\ B_2H_6 + 3\ LiAlF4 + 4\ Et_2O$. Redox Prep: $2\ NaBH4 + I_2 \rightarrow B_2H_6 + 2\ NaI + H_2$ (in diglyme).

3. Reactions of B2H6

- **Combustion:** $B_2H_6 + 3\ O_2 \rightarrow B_2O_3 + 3\ H_2O$ ($\Delta H° = -1976$ kJ/mol, violent green flame, high fuel value).
- **Hydrolysis:** $B_2H_6 + 6\ H_2O \rightarrow 2\ H_3BO_3 + 6\ H_2$ (rapid, quantitative evolution of hydrogen).
- **Hydroboration:** $B_2H_6 + 6\ RCH{=}CH2 \rightarrow 2\ B(CH_2CH_2R)_3$ (syn-addition, anti-Markovnikov alcohol precursor).

Higher Boranes & Hydroborates — Wade's Rules, NaBH4 & Borazine

Boranes form extensive cluster families classified by skeletal electron pairs (Wade-Mingos rules): Closo ($B_nH_{n+2}^{2-}$), Nido (B_nH_{n+4}), Arachno (B_nH_{n+6}), and Hypho (B_nH_{n+8}).

1. Borazine B3N3H6 ('Inorganic Benzene')

Heating the unsymmetrical cleavage product of diborane and ammonia gives Borazine: $3\ B_2H_6 + 6\ NH_3$ (low T) → $3\ [H_2B(NH_3)_2]^+[BH_4]^-$ --(heat, 450 K)--→ **$2\ B_3N_3H_6$** + $12\ H_2$ Alternate synthesis: $3\ BCl_3 + 3\ NH_4Cl \rightarrow (ClBNH)_3 + 9\ HCl$; reduce with $NaBH_4 \rightarrow B_3N_3H_6$.

Property	Benzene (C_6H_6)	Borazine ($B_3N_3H_6$)
Aromaticity	Completely delocalised (150 kJ/mol)	Partially delocalised (pi polar towards N)
Polarity & Reactivity	Non-polar, electrophilic substitution	Polar ($B^{\delta+}$ - $^{N}\delta-$), addition reactions dominant
Addition of HCl	No reaction without $AlCl_3$	Adds 3 HCl readily: H to N, Cl to B → $B_3N_3H_9Cl_3$
Hydrolysis	Resistant to hydrolysis	Hydrolyses readily: $B_3N_3H_6 + 9\ H_2O \rightarrow 3\ H_3BO_3 + 3\ NH_3 + 3\ H_2$

Table 13.3 — Comparison of Benzene vs Inorganic Benzene.

KEY RESULT
Borazine is **iso-electronic and iso-structural** to Benzene (planar hexagonal ring, sp^2 B and N), but chemically MUCH more reactive due to electronegativity difference ($\chi_N = 3.0$, $\chi_B = 2.0$).

2. Sodium Borohydride (NaBH4) & LiAlH4

- **NaBH4 Synthesis:** $4\ NaH + B(OMe)_3 \rightarrow NaBH_4 + 3\ NaOMe$ (at 250°C). Structure: Ionic $[Na]^+\ [BH_4]^-$ (tetrahedral BH_4^-, sp^3 B, regular 2c-2e B-H bonds, 8 valence e_^).
- **LiAlH4 Synthesis:** $4\ LiH + AlCl_3 \rightarrow LiAlH_4 + 3\ LiCl$ (in dry ether). Reducing power: $LiAlH_4$ is a far stronger hydride donor than $NaBH_4$ because Al-H is more polar and weaker than B-H. $LiAlH_4$ reacts violently with water/alcohols; $NaBH_4$ can be used in basic water/MeOH.

TRAP
TRAP: $NaBH_4$ reduces aldehydes and ketones to alcohols, but does NOT reduce esters, amides, or carboxylic acids under standard conditions. $LiAlH_4$ reduces all of them.

Boron Trihalides (BX3) — pπ-pπ Back-Bonding & Lewis Acidity Inversion

In BX3 (X = F, Cl, Br, I), Boron is sp^2 hybridized with a vacant, unhybridized 2p_z orbital perpendicular to the planar molecule. The halogen atom has filled p orbitals with lone pairs.

1. pπ-pπ Back-Bonding Mechanism

The lone pair in a filled 2p orbital of Fluorine overlaps sideways with the empty 2p orbital of Boron, forming a **pπ-pπ dative back-bond**. This gives partial double bond character to the B-X bond.

Halide	Back-Bond Overlap	Extent of Back-Bonding	Observed B-X Bond Length
BF3	2p(B) <--- 2p(F) (identical size, excellent match)	Maximum back-bonding	130 pm (much shorter than single bond 152 pm)
BCl3	2p(B) <--- 3p(Cl) (diffuse 3p, poor overlap)	Weak back-bonding	175 pm
BBr3	2p(B) <--- 4p(Br) (very diffuse 4p)	Very weak back-bonding	187 pm
BI3	2p(B) <--- 5p(I) (energy/size mismatch)	Negligible back-bonding	210 pm

Table 13.4 — Back-bonding variation across Boron Trihalides.

KEY RESULT

LEWIS ACIDITY ORDER: **BF3 < BCl3 < BBr3 < BI3** (Opposite of halogen electronegativity!). Reason: In BF3, strong 2p-2p back-bonding satisfies Boron's electron deficiency. In BI3, negligible back-bonding leaves Boron completely vacant and most hungry for external electron pairs.

2. Hydrolysis of Boron Trihalides

- **BCl3, BBr3, BI3:** Undergo complete hydrolysis without secondary complexation: $BX_3 + 3\ H_2O \rightarrow H_3BO_3 + 3\ HX$ (vigorous, fumes in moist air).
- **BF3 Anomaly (Partial Hydrolysis):** Step 1: $4\ BF_3 + 3\ H_2O \rightarrow B(OH)_3 + 3\ HBF_4$ (or: $BF_3 + 3\ H_2O \rightarrow H_3BO_3 + 3\ HF$) Step 2: HF immediately attacks unreacted BF3: $3\ BF_3 + 3\ HF \rightarrow$ **$3\ H^+\ [BF_4]^-$** (Fluoroboric acid). Net reaction: **$4\ BF_3 + 3\ H_2O \rightarrow H_3BO_3 + 3\ H[BF_4]$** (does not go to completion to yield pure HF!).

TRAP

TRAP: BF3 does NOT give white fumes of HF on hydrolysis because HF reacts with BF3 to form HBF4! BCl3 fumes copiously because HCl gas does not coordinate to BCl3.

Orthoboric Acid & Borax — Layered Hydrogen Bonding & Aprotic Acidity

Orthoboric acid H_3BO_3 (or $B(OH)_3$) is a white, slippery, crystalline solid with a layered sheet structure held together by extensive planar hydrogen bonds. It acts as a mild antiseptic.

1. Nature of Acidity: Monobasic Lewis Acid

H_3BO_3 is **NOT a protonic Bronsted acid** (does not lose its own H^+). It acts as a **monobasic Lewis acid** by accepting OH^- from water, thereby releasing H^+ from the solvent:

KEY RESULT
$B(OH)_3 + 2\ H_2O \rightleftharpoons [B(OH)_4]^- + H_3O^+$ ($pK_a = 9.25$, very weak) It cannot be titrated directly against standard NaOH because the endpoint is diffuse.

2. Enhancement of Acidity by Polyhydroxy Compounds (cis-Diols)

When cis-diols (glycerol, mannitol, catechol, or sorbitol) are added, the $[B(OH)_4]^-$ chelate forms a very stable spirocyclic bis-chelate complex, shifting the equilibrium completely to the right:

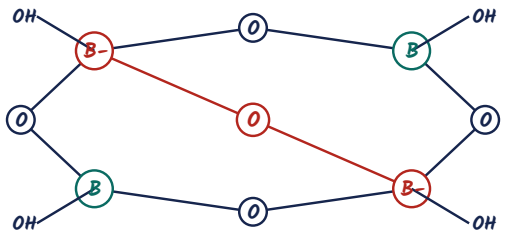
- Reaction: $B(OH)_3 + 2\ C_3H_5(OH)_3 \rightleftharpoons [B(C_3H_6O_3)_2]^- + H_3O^+ + 2\ H_2O$.
- Result: Boric acid becomes a **strong monoprotic acid** and can be titrated sharply against NaOH using phenolphthalein indicator!
- Requirement: The diol must have **cis-vicinal -OH groups** (trans-diols do not enhance acidity).

3. Thermal Action on Boric Acid

Stepwise dehydration on heating: H_3BO_3 --(373 K, $-H_2O$)--→ **HBO2 (Metaboric acid)** --(433 K, $-H_2O$)--→ **$H_2B_4O_7$ (Tetraboric acid)** --(red heat)--→ **B_2O_3 (Boron sesquioxide / glassy mass)**.

4. Borax (Na2B4O7·10H2O)

Correct structural formula: **$Na_2[B_4O_5(OH)_4]\cdot 8H2O$**. It contains 8 water molecules of crystallization.



[B4O5(OH)4]2- : Two sp2 B (planar) + Two sp3 B- (tetrahedral) + 5 B-O-B bridges

Fig. 13.2 — Structure of the $[B_4O_5(OH)_4]2-$ anion in Borax.

- **Borax in Water:** Undergoes hydrolysis to give an alkaline buffer solution: $Na_2B_4O_7 + 7\ H_2O \rightarrow 2\ NaOH + 4\ H_3BO_3$ (solution is alkaline, pH ≈ 9).

Borax Bead Test & Chemical Analysis — Colored Metaborates & Fluoroborates

On heating in a platinum wire loop, borax swells up as water of crystallization is lost, and then melts into a clear, transparent, glassy bead containing **$NaBO2 + B_2O_3$**.

1. Chemistry of the Borax Bead Test

$Na_2B_4O_7·10H_2O$ --(heat, -10 H_2O)--→ $Na_2B_4O_7$ --(fuse)--→ **2 NaBO2 (Sodium metaborate) + B_2O_3 (Boric anhydride)** When heated with transition metal salts, B_2O_3 reacts with metal oxides to form characteristically coloured **metaborates $M(BO_2)_2$**:

Metal Cation	Oxidising Flame (Hot / Cold)	Reducing Flame (Hot / Cold)	Metaborate Formed
Cu^{2+}	Green (hot) / Blue (cold)	Colorless (hot) / Opaque Red (cold, Cu metal)	$Cu(BO_2)_2$ / Cu
Co^{2+}	Deep Blue (hot & cold)	Deep Blue (hot & cold)	$Co(BO_2)_2$
Cr^{3+}	Dark Yellow (hot) / Green (cold)	Green (hot & cold)	$Cr(BO_2)_3$
Fe^{3+}	Yellowish-brown (cold)	Bottle Green (cold)	$Fe(BO_2)_2$ / $Fe(BO_2)_3$
Ni^{2+}	Violet/Brown (hot) / Brown (cold)	Grey (reduced Ni particles)	$Ni(BO_2)_2$

Table 13.5 — Characteristic flame colours in Borax Bead Test.

2. Qualitative Test for Borate: Ethyl Borate Test

Borate salt + conc. H_2SO_4 + Ethanol → Heat → **Triethyl borate $B(OC_2H_5)_3$** vapours burn with a characteristic **green-edged flame**: H_3BO_3 + 3 C_2H_5OH --(conc. H_2SO_4)--→ $B(OC_2H_5)_3$↑ + 3 H_2O.

3. Fluoroboric Acid (HBF4)

- Preparation: $B(OH)_3$ + 4 HF → $H[BF4]$ + 3 H_2O.
- Structure: $[BF4]^-$ is perfectly tetrahedral (sp^3, 4 equivalent B-F bonds of 143 pm).
- Properties: Strong acid, non-coordinating anion, used in Balz-Schiemann diazotisation ($ArN2^+$ $BF4^-$ → ArF + N_2 + BF3).

Boron Nitride $(BN)_x$ — Synthesis, Allotropes & Comparison with Carbon

Boron Nitride is iso-electronic with Carbon (B + N = 3 + 5 = 8 e^- = 2 × C). It exists in two major crystalline forms: Hexagonal BN (h-BN, inorganic graphite) and Cubic BN (c-BN, borazon, inorganic diamond).

1. Hexagonal Boron Nitride (h-BN: 'Inorganic Graphite')

- **Synthesis:** $B_2O_3 + 2\ NH_3 \xrightarrow{(1200\ K)} 2\ BN + 3\ H_2O$ (or: $BCl_3 + NH_3 \rightarrow BN + 3\ HCl$ at 1000°C).
- **Structure:** Planar hexagonal rings of alternating B and N atoms (B-N bond length 145 pm).
- **Layer Stacking Difference:** In graphite, layers are stacked in an ABAB pattern with atoms displaced. In h-BN, layers are stacked directly **AA'AA'** with B atoms directly over N atoms in adjacent layers (dipole-dipole attraction $B^{\delta+\cdots N}\delta-$).
- **Electrical Conductivity:** Graphite is a good conductor (delocalised π electrons). h-BN is a **white insulator** (wide bandgap ≈ 5.9 eV) because π electrons are tightly held by electronegative Nitrogen atoms!

Feature	Graphite (Carbon)	Hexagonal Boron Nitride (h-BN)
Color & Appearance	Grey-black, metallic luster	Pure white solid ('White Graphite')
Layer Stacking	ABAB... (staggered rings)	AA'AA'... (B above N eclipsed)
Electrical Conduction	Conductor along planes (semimetal)	Electrical Insulator (wide bandgap)
Chemical Resistance	Oxidises at 700°C	Resistant to oxidation up to 1000°C, inert to acids

Table 13.6 — Direct structural comparison between Graphite and h-BN.

2. Cubic Boron Nitride (c-BN: 'Borazon')

Formed by heating h-BN under extreme pressure (60,000 atm, 1800°C). It has a zinc-blende (diamond-like) three-dimensional tetrahedral network. It is second only to diamond in hardness, but has higher thermal stability in air and does not react with ferrous metals during high-speed cutting.

Aluminium Chemistry — Halide Dimers, Alums, Passivation & Thermite

Aluminium is the most abundant metal in the Earth's crust (8.3% by mass). Key ore is **Bauxite** **AlOx(OH)3-2x** purified by Bayer's process and reduced by the Hall-Héroult electrolytic process.

1. Aluminium Chloride (AlCl3 vs Al2Cl6)

- **Anhydrous AlCl3:** White crystalline solid, highly hygroscopic. Prepared by: $2\ Al + 3\ Cl_2 \rightarrow 2\ AlCl3$ (or: $Al_2O_3 + 3\ C + 3\ Cl_2 \rightarrow 2\ AlCl3 + 3\ CO$).
- **Dimerisation in Vapor/Non-polar solvents:** AlCl3 dimerises to **Al_2Cl_6** through two bridging dative bonds (Cl: → Al). Each Al is sp^3 hybridized with an octet.
- **High Temperature Dissociation:** At T > 1000 K, Al_2Cl_6 dissociates into planar monomeric **AlCl3** (sp^2, Lewis acid).
- **In Water:** Al_2Cl_6 dissolves exothermically and ionizes: $Al_2Cl_6 + 12\ H_2O \rightarrow 2\ [Al(H_2O)_6]^{3+} + 6\ Cl^-$ (octahedral complex, acidic pH due to hydrolysis).

KEY RESULT
AlCl3 cannot be prepared anhydrous by heating AlCl3·6H2O! On heating, it undergoes hydrolysis: 2 AlCl3·6H2O --(heat)--→ $Al_2O_3 + 6\ HCl + 9\ H_2O$.

2. Alums: General Formula M2SO4·M'2(SO4)3·24H2O

Where M = univalent cation (K^+, Na^+, $NH4^+$, Rb^+, Cs^+; NOT Li^+ due to small size) and M' = trivalent cation (Al^{3+}, Cr^{3+}, Fe^{3+}, Co^{3+}). Structural formula: **$[M(H_2O)_6]^+\ [M'(H_2O)_6]^{3+}\ (SO4^{2-})2$**.

Common Alum	Formula	Primary Uses
Potash Alum	$K_2SO_4·Al_2(SO_4)_3·24H_2O$	Water purification (coagulation of colloidal clay), mordant in dyeing
Ammonium Alum	$(NH4)2SO4·Al_2(SO_4)_3·24H_2O$	Baking powder, water treatment, fire-retardant textiles
Chrome Alum	$K_2SO_4·Cr_2(SO_4)_3·24H_2O$	Leather tanning (cross-links collagen proteins)
Ferric Alum	$(NH4)2SO4·Fe_2(SO_4)_3·24H_2O$	Indicator in Volhard's titration of halides

Table 13.7 — Composition and applications of Alums in JEE.

3. Passivation & Goldschmidt Thermite Process

- **Passivation:** Conc. HNO3 makes Al passive by forming a dense, insoluble, protective oxide film of Al_2O_3. Hence, conc. HNO3 can be transported in Al containers.
- **Thermite Reaction:** $Fe_2O_3 + 2\ Al \rightarrow Al_2O_3 + 2\ Fe(l)$ (ΔH = -850 kJ/mol, T ≈ 3000°C, used for on-site welding of railway tracks).

Group 14: Carbon Family ($ns^2\ np2$) — Periodic Trends, Catenation & Inert Pair

Contains C (non-metal), Si (metalloid), Ge (metalloid), Sn and Pb (metals). Key characteristics: remarkable catenation ability of Carbon, pπ-pπ vs pπ-dπ multiple bonding, and strong inert pair effect making Pb(II) far more stable than Pb(IV).

1. Periodic Trends in Group 14

Element	Config.	Covalent Radius (pm)	IE-1 (kJ/mol)	M.P. (°C)	Electronegativity
C	[He] $2s^2\ 2p2$	77	1086	4100 (diamond)	2.5
Si	[Ne] $3s^2\ 3p2$	118	786	1410	1.8
Ge	[Ar] $3d^{10}\ 4s2$ 4p^2	122	761	937	1.8
Sn	[Kr] $4d^{10}\ 5s2$ 5p^2	140	708	232	1.8
Pb	[Xe] $4f^{14}\ 5d10\ 6s^2\ 6p2$	146	715 (Pb > Sn!)	327	1.9

Table 14.1 — Periodic properties of Group 14 elements.

KEY RESULT
Ionization Enthalpy: **C > Si > Ge > Pb > Sn** (Pb has higher IE than Sn due to poor 4f shielding).
Catenation Order: **C >>> Si > Ge ≈ Sn >> Pb** (Directly follows E-E single bond strength: C-C 348 > Si-Si 297 > Ge-Ge 260 > Sn-Sn 240 kJ/mol).

2. Inert Pair Effect & Redox Behaviour of Sn and Pb

- **+4 Oxidation State:** C(+4), Si(+4), Ge(+4), Sn(+4) are stable. Pb(+4) is unstable and a **powerful oxidising agent** (PbO2 + 4 HCl → PbCl2 + Cl_2 + 2 H_2O).
- **+2 Oxidation State:** Sn(+2) is a **strong reducing agent** ($Sn^{2+} \rightarrow Sn^{4+} + 2e^-$, $E° = -0.15$ V). Used to reduce FeCl3 to FeCl2, and HgCl2 to white Hg_2Cl_2 then black Hg.
- **Pb(+2) vs Pb(+4):** Pb(+2) is extremely stable. PbCl4 decomposes on warming to PbCl2 + Cl_2. PbI4 and PbBr4 do not even exist because I^- and Br^- reduce Pb^{4+} instantly!

TRAP
TRAP: PbO2 dissolves in conc. HNO3 with evolution of O_2: 2 PbO2 + 4 HNO3 → 2 $Pb(NO_3)_2$ + 2 H_2O + $O_2\uparrow$. It does NOT form a nitrate of lead(IV)!

3. Multiple Bonding: pπ-pπ vs dπ-pπ

Carbon forms very strong **pπ-pπ multiple bonds** (C=C, C≡C, C=O, C≡N) because 2p orbitals are compact and overlap sideways efficiently. Heavier elements (Si, Ge, Sn) have large, diffuse p orbitals that cannot form stable pπ-pπ bonds; they prefer single bonds or use empty d orbitals (dπ-pπ).

Allotropes of Carbon — Diamond, Graphite, Fullerenes & Thermodynamic Stability

Carbon exhibits rich allotropy due to catenation and ability to adopt sp^3, sp^2, and sp hybridization.

1. Structural Comparison of Major Allotropes

Property	Diamond	Graphite	Fullerene (C_{60})	
Hybridization & Geometry	sp^3, 3D tetrahedral network	sp^2, 2D planar hexagonal sheets	sp^2 (curved), soccer-ball (Ih symmetry)	
C-C Bond Length	154 pm (all uniform single bonds)	141.5 pm (in-plane), 340 pm (inter-layer)	145 pm (single, 6-5 ring), 138 pm (double, 6-6 ring)	
Conductivity	Electrical insulator (bandgap 5.5 eV)	Good electrical conductor along planes	Semiconductor; K_3C_{60} is a superconductor (18 K)	
Density & Hardness	3.51 g/cm^3, Hardest natural substance	2.22 g/cm^3, Soft, slippery (solid lubricant)	1.65 g/cm^3, Molecular solid (soluble in toluene)	
Standard Heat of Formation	+1.90 kJ/mol (metastable at RTP)	**0.00 kJ/mol (Thermodynamically MOST stable)**	+38.1 kJ/mol	

Table 14.2 — Comparative crystallography and physics of Carbon allotropes.

KEY RESULT

Thermodynamic Stability: Graphite is the standard state of carbon ($\Delta H°_f = 0$). Diamond conversion to graphite is thermodynamically favored ($\Delta G < 0$) but kinetically dormant (huge activation energy).

2. Buckminsterfullerene (C60) Architecture

- **Faces & Rings:** C_{60} contains **20 six-membered rings (hexagons)** and **12 five-membered rings (pentagons)**.
- **Isolated Pentagon Rule (IPR):** Six-membered rings can fuse with both hexagons and pentagons, but **five-membered rings ONLY fuse with six-membered rings** (no two pentagons share an edge, minimising strain).
- **Bond Alternation:** C=C double bonds exist strictly at the junction of two hexagons (6-6 ring fusion, length 138 pm). Single bonds exist at 6-5 junctions (145 pm).

3. Amorphous Carbon Allotropes & Adsorption

Coke, Charcoal, Lampblack, Carbon black, and Activated Charcoal. Activated charcoal has an immense surface area (~1000 m^2/g) and is used in gas masks, decolourisation of sugar, and water filtration.

Carbides — Ionic (Salt-like), Covalent, Interstitial & Hydrolysis Pathways

Carbides are binary compounds of carbon with elements of lower or comparable electronegativity. They are classified into three distinct structural classes based on bonding.

1. Salt-like (Ionic) Carbides & Hydrolysis Products

Formed by highly electropositive metals (Group 1, 2, Al). Classified by the hydrocarbon gas evolved on hydrolysis:

Carbide Type	Carbon Anion	Example Carbides	Hydrolysis Reaction & Gas Evolved
Methanides	C^{4-} (isolated carbon)	Al_4C_3, Be_2C	$Al_4C_3 + 12\ H_2O \rightarrow 4\ Al(OH)_3 + \mathbf{3\ CH_4}\uparrow$ (Methane)
Acetylides	C_2^{2-} ($[:C\equiv C:]^{2-}$)	CaC2, MgC2, Cu_2C_2, Ag_2C_2	$CaC2 + 2\ H_2O \rightarrow Ca(OH)_2 + \mathbf{C_2H_2}\uparrow$ (Ethyne / Acetylene)
Allylenides	C_3^{4-} ($[:C=C=C:]^{4-}$)	Mg_2C_3	$Mg_2C_3 + 4\ H_2O \rightarrow 2\ Mg(OH)_2 + \mathbf{CH3\text{-}C\equiv CH}\uparrow$ (Propyne / Allylene)

Table 14.3 — Ionic carbides and their diagnostic hydrolysis gases (JEE Favourite).

KEY RESULT
Memory Rule for Hydrolysis: · **Be_2C & Al_4C_3** → CH_4 (Methanides) · **CaC2, BaC2, Na_2C_2** → C_2H_2 (Acetylides) · **Mg_2C_3** → CH3-C≡CH (Allylenide / Propyne)

2. Covalent & Interstitial Carbides

- **Covalent Carbides:** Formed by elements of similar size and electronegativity: · **Silicon Carbide (SiC, 'Carborundum'):** 3D diamond-like covalent network. Extremely hard (Mohs 9.5), inert to acids, refractory abrasive (prep: SiO2 + 3 C --(2000°C)--> SiC + 2 CO). · **Boron Carbide (B_4C):** Hardest artificial carbide, used in tank armor and nuclear reactor control rods.
- **Interstitial Carbides:** Formed by transition metals (TiC, WC, TaC, V_4C_3, Fe_3C): Small carbon atoms occupy octahedral interstitial voids in the close-packed metal lattice without changing metal structure. Retain metallic electrical conductivity, possess ultra-high melting points (>3000°C), and extreme hardness.

Oxides of Carbon — CO (Ligand & Toxicity), CO2, C_3O_2 & Carbonate Chemistry

Carbon forms three well-characterised oxides: neutral CO, acidic CO2, and carbon suboxide C_3O_2.

1. Carbon Monoxide (CO) — Structure & Bonding

Iso-electronic with N_2 and CN^- (10 valence electrons). Molecular Orbital configuration: σ(2s)² σ*(2s)2 π(2p_x)² π(2p_y)2 σ(2p_z)^2. Bond order = 3 (one sigma, two pi bonds).

- **Dipole Moment:** Remarkably low (0.11 D) directed $C^{\delta-} \leftarrow\!\!- {}^{O}\delta+$ (dative lone pair from O to C cancels electronegativity difference!).
- **Coordination Chemistry:** Acts as a strong **π-acid ligand** in transition metal carbonyls (e.g. $Ni(CO)_4$, $Fe(CO)_5$) via σ-donation from C and π-backbonding into empty π orbitals of CO.
- **Toxicity:** Binds to blood haemoglobin 300 times more strongly than O2~ to form **carboxyhaemoglobin**, preventing oxygen delivery.

Laboratory Synthesis	Industrial / Gas Mixtures	Reactions
Dehydration of Formic Acid: HCOOH --(conc. H_2SO_4, 373 K)--> CO + H_2O	Water Gas (Synthesis gas): C + $H_2O(g)$ --(1270 K)--> CO + H_2	Combustion: 2 CO + O_2 → 2 CO2 (blue flame)
Dehydration of Oxalic Acid: (COOH)2 --(conc. H_2SO_4)--> CO + CO2 + H_2O (absorb CO2 in KOH)	Producer Gas: 2 C + O_2 + 4 N_2 --(1270 K)--> 2 CO + 4 N_2	Mond Process: Ni + 4 CO --(330 K)--> $Ni(CO)_4$ --(450 K)--> Ni + 4 CO

Table 14.4 — Synthesis and reactions of Carbon Monoxide.

2. Carbon Suboxide (C3O2)

Prepared by dehydrating malonic acid with phosphorus pentoxide: CH2(COOH)2 + P_4O_{10} --(150°C)--> **O=C=C=C=O** + 4 HPO3 Structure: Linear, cumulative double bond system, sp-hybridized central carbons, bp 7°C, foul odor. Hydrolyses back to malonic acid.

3. Carbon Dioxide (CO2) & Bicarbonate Buffers

- Linear molecule (O=C=O, $D_{\infty h}$ symmetry), μ = 0, C-O bond length 116 pm (partial triple bond character).
- Lime water test: $Ca(OH)_2$ + CO2 → **CaCO3↓** (white turbidity) + H_2O. Excess CO2 dissolves turbidity: CaCO3 + CO2 + H_2O → **$Ca(HCO_3)_2$** (soluble). On boiling, turbidity returns!

Silicon Dioxide (SiO2) & Silanes — 3D Network Structure & Hydrolysis

While CO2 is a discrete linear gas, SiO2 is a high-melting 3D network solid because Si cannot form stable pπ-pπ double bonds with oxygen.

1. Structure & Allotropy of Silica (SiO2)

- **Network Architecture:** Every Si atom is **sp^3 hybridized** and tetrahedrally bonded to 4 Oxygen atoms. Every O atom is shared between 2 Si atoms (Si-O-Si angle ≈ 144°), yielding empirical formula SiO2.
- **Crystalline Forms:** Quartz (piezoelectric), Tridymite, and Cristobalite (interconvertible at high temperatures).
- **Chemical Inertness:** Very inert; attacked only by HF and strong hot caustic alkali: · With HF (Etching of glass): $SiO_2 + 4\ HF \rightarrow \mathbf{SiF_4} + 2\ H_2O$; $SiF_4 + 2\ HF \rightarrow \mathbf{H_2SiF_6}$ (Hydrofluorosilicic acid). · With NaOH: $SiO_2 + 2\ NaOH \rightarrow \mathbf{Na_2SiO_3}$ (Water glass) + H_2O.

2. Silanes (Si_nH_{2n+2}) vs Alkanes

Silanes are the silicon hydrides (SiH_4, Si_2H_6, up to Si_6H_{14}), prepared by reacting Mg_2Si with dil. HCl: $Mg_2Si + 4\ HCl \rightarrow 2\ MgCl_2 + SiH_4$ (Monosilane).

Property	Methane (CH_4)	Silane (SiH_4)
Combustion / Air Stability	Stable in air, requires ignition ($\Delta H_{comb} = -890$ kJ)	Spontaneously inflammable in air (pyrophoric!): $SiH_4 + 2\ O_2 \rightarrow SiO_2 + 2\ H_2O$
Hydrolysis by Water	Completely inert to water and aqueous alkali	Violently hydrolysed by even traces of base: $SiH_4 + (n+2)\ H_2O \xrightarrow{(OH^-)} SiO_2{\cdot}nH_2O + 4\ H_2\uparrow$
Polarity of E-H Bond	$C^{\delta-} - H^{\delta+}$ ($\chi_C = 2.5 > \chi_H = 2.1$)	$Si^{\delta+} - H^{\delta-}$ ($\chi_{Si} = 1.8 < \chi_H = 2.1$; Hydridic H!)
Mechanism of Attack	Carbon has no vacant d orbitals; SN2 hindered	Si has accessible empty 3d orbitals; nucleophilic attack by OH^- is very fast

Table 14.5 — Why Silane is hyper-reactive compared to Methane.

TRAP

TRAP: Carbon tetrachloride CCl_4 is completely inert to water, but Silicon tetrachloride $SiCl_4$ hydrolyses violently: $SiCl_4 + 4\ H_2O \rightarrow Si(OH)_4 + 4\ HCl$. Si utilizes vacant 3d orbitals to expand its coordination number!

Silicate Minerals — Classification, Structural Units & Sharing of Oxygen Vertices

The basic building unit of all silicates is the tetrahedral **$[SiO_4]^{4-}$** unit. Silicates are classified strictly by how many corners (oxygen atoms) are shared between adjacent $[SiO_4]$ tetrahedra.

1. Classification of Silicates

Silicate Class	Shared O / Tetrahedron	General Formula Unit	Mineral Examples
1. Orthosilicates (Neso)	0 (discrete units)	$[SiO_4]^{4-}$	Phenacite (Be_2SiO_4), Willemite (Zn_2SiO_4), Olivine (Mg_2SiO_4), Forsterite
2. Pyrosilicates (Soro)	1 (dimeric unit)	$[Si_2O_7]^{6-}$	Thortveitite ($Sc_2Si_2O_7$), Hemimorphite ($Zn_4(OH)_2Si_2O_7·H_2O$)
3. Cyclic / Ring	2 (closed rings)	$[Si_3O_9]^{6-}$ or $[Si_6O_{18}]^{12-}$	Benitoite ($BaTiSi_3O_9$), Beryl ($Be_3Al_2Si_6O_{18}$ — gemstone emerald)
4. Single Chain (Ino / Pyroxenes)	2 (infinite line)	$([SiO_3]^{2-})_n$	Diopside ($CaMg(SiO_3)_2$), Spodumene ($LiAl(SiO_3)_2$), Enstatite
5. Double Chain (Amphiboles)	2.5 (alternating 2 and 3)	$([Si_4O_{11}]^{6-})_n$	Tremolite ($Ca_2Mg_5(Si_4O_{11})_2(OH)_2$), Asbestos minerals
6. Sheet Silicates (Phyllo)	3 (2D planar infinite sheets)	$([Si_2O_5]^{2-})_n$	Talc ($Mg_3(Si_2O_5)_2(OH)_2$), Kaolinite (clay), Mica, Muscovite
7. 3D Framework (Tecto)	4 (all four corners shared)	$(SiO_2)_n$	Quartz, Feldspar, Zeolites, Ultramarines

Table 14.6 — Master classification of Silicate minerals (JEE Advanced core).

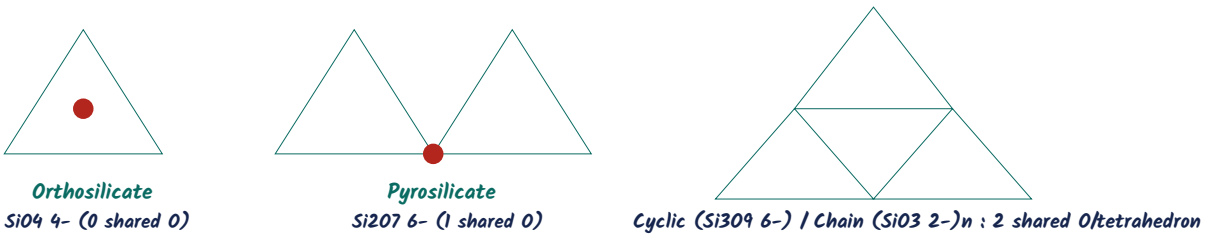


Fig. 14.1 — Structural polyhedra of Ortho, Pyro, and Cyclic silicates.

KEY RESULT

General Charge Rule: Formula of a silicate sharing 'c' corners per tetrahedron is **$[SiO_{4-c/2}]^{2(4-c)-}$**. · c = 0 → $[SiO_4]^{4-}$ (Ortho) · c = 1 → $[Si_2O_7]^{6-}$ (Pyro) · c = 2 → $[SiO_3]_n^{2n-}$ (Cyclic & Single Chain) · c = 3 → $[Si_2O_5]_n^{2n-}$ (Sheet) · c = 4 → SiO_2 (3D Framework)

Silicones & Zeolites — Synthetic Polymers & Shape-Selective Catalysts

Silicones are synthetic organosilicon polymers containing repeating **$-R_2Si-O-$** units. Zeolites are microporous aluminosilicate 3D framework molecular sieves.

1. Synthesis & Control of Silicones

Prepared from alkylchlorosilanes synthesized by Rochow reaction: $2\ RCl + Si \xrightarrow{Cu\ powder,\ 570\ K} R_2SiCl_2$:

- **Chain Propagation (Linear Silicones):** Hydrolysis of **R_2SiCl_2** gives $R_2Si(OH)_2$ which undergoes linear condensation polymerization: $n\ R_2SiCl_2 + 2n\ H_2O \rightarrow n\ R_2Si(OH)_2 \rightarrow$ **...-O-Si(R_2)-O-Si(R_2)-O-...** (linear silicone).
- **Chain Termination (Controlling Molecular Weight):** Addition of **R_3SiCl** produces monofunctional R_3Si-OH, capping the chain ends and terminating growth.
- **Cross-Linked (Resin) Silicones:** Hydrolysis of **RSiCl3** produces trifunctional RSi(OH)3 which condenses in three directions forming a rigid 3D cross-linked silicone polymer.

Monomer Precursor	Hydrolysis Intermediate	Resulting Silicone Polymer Structure
R_3SiCl (monochlorosilane)	R_3SiOH (monofunctional)	Dimer: $R_3Si-O-SiR_3$ (Chain Stopper / Capping agent)
R_2SiCl_2 (dichlorosilane)	$R_2Si(OH)_2$ (difunctional)	Linear chain polymer or Cyclic oligomers (e.g. $(R_2SiO)_3$)
RSiCl3 (trichlorosilane)	RSi(OH)3 (trifunctional)	Cross-linked 3D network resin (high thermal rigidity)

Table 14.7 — Role of chlorosilane precursors in silicone architecture.

2. Properties & Applications of Silicones

High thermal stability, water-repellent (hydrophobic organic R groups point outward), electrical insulators, biocompatible (prosthetics, lubricants, waterproof fabrics).

3. Zeolites: Three-Dimensional Aluminosilicates

- **Formula & Charge Balance:** In 3D framework SiO2, when some Si^{4+} atoms are replaced by Al^{3+}, the framework acquires a net negative charge balanced by extra-framework cations (Na^+, K^+, Ca^{2+}): **M_{x/n} [(AlO2)_x (SiO2)_y]·z H_2O**.
- **Shape-Selective Catalysis:** Zeolites possess open honeycomb cages with uniform pore sizes (3 to 10 Å). Only molecules of fitting dimensions can enter and react.
- **ZSM-5 (Zeolite Socony Mobil-5):** Converts alcohols directly into gasoline (hydrocarbons) by shape-selective dehydration: **x CH3OH --(ZSM-5, 670 K)--> Gasoline ($C_n H_{2n+2}$) + x H_2O**.
- **Water Softening:** Permutit (hydrated sodium aluminosilicate $Na_2Al_2Si_2O_8 \cdot xH_2O$) exchanges Na^+ for Ca^{2+} and Mg^{2+} in hard water.

Group 14 Halides & Redox Trends — Tetrahalides vs Dihalides

Tetrahalides MX4 (sp^3 tetrahedral) dominate for C, Si, Ge, and Sn, whereas dihalides MX2 (bent/polymeric) become progressively stable descending to Pb due to the inert pair effect.

1. Tetrahalides: Hydrolysis Mechanisms & Exceptions

- **CCl4 Inertness:** CCl4 does not hydrolyse in neutral or basic water because Carbon is in the 2nd period: no vacant d orbitals are available to expand its octet to accept the nucleophilic lone pair of water.
- **SiCl4, GeCl4, SnCl4 Hydrolysis:** Rapid and exothermic. Water coordinates to vacant d orbitals of the central atom forming a 5-coordinate intermediate, followed by elimination of HCl: SiCl4 + 4 H_2O → $[Si(OH)_4]$ (orthosilicic acid) → SiO2·xH2O + 4 HCl.
- **Lead Tetrahalides:** PbCl4 is a yellow oily liquid that readily decomposes to PbCl2 + Cl_2. **PbBr4 and PbI4 do not exist** because Pb^{4+} oxidises Br^- and I^- spontaneously.

Halide	Physical State	Oxidation State	Key Chemical Reaction / Behaviour
SnCl2	White solid	+2 (reducing)	SnCl2 + 2 FeCl3 → SnCl4 + 2 FeCl2; reduces HgCl2 → $Hg_2Cl_2\downarrow$ → Hg↓
SnCl4	Colorless liquid (fuming)	+4	Forms hexachlorostannate with HCl: SnCl4 + 2 HCl → $H_2[SnCl6]$
PbCl2	White crystalline solid	+2 (stable)	Insoluble in cold water, soluble in hot water (recrystallises as needles)
PbCl4	Yellow liquid (unstable)	+4 (oxidising)	Decomposes explosively on heating: PbCl4 --(Δ)--→ PbCl2 + $Cl_2\uparrow$

Table 14.8 — Comparative chemistry of Tin and Lead chlorides.

2. Dihalides & Carbene Analogues

In the gas phase, Group 14 dihalides (CF2, SiCl2, GeCl2, SnCl2, PbCl2) are VSEPR **bent (V-shaped)** molecules with a lone pair and an empty p orbital (carbene analogues). In the solid state, SnCl2 and PbCl2 form extended halogen-bridged polymers or ionic lattices.

KEY RESULT

Oxidising vs Reducing Summary: · **Sn^{2+} is REDUCING:** $Sn^{2+} \rightarrow Sn^{4+} + 2e^-$ (favors +4 state) · **Pb^{4+} is OXIDISING:** $Pb^{4+} + 2e^- \rightarrow Pb^{2+}$ (favors +2 state due to inert pair effect)

Group 15: Nitrogen Family ($ns^2\ np^3$) — Pnictogens & Periodic Trends

Contains N, P (non-metals), As, Sb (metalloids), Bi (post-transition metal). The valence shell has a stable half-filled p^3 configuration, resulting in unusually high ionization energies.

1. Periodic Trends of Group 15

Element	Config.	Covalent Radius (pm)	IE-1 (kJ/mol)	Electronegativity	Allotropes
N	[He] $2s^2\ 2p^3$	70	1402	3.0	Dinitrogen gas (α, β solid forms)
P	[Ne] $3s^2\ 3p^3$	110	1012	2.1	White, Red, Black (α, β), Violet
As	[Ar] $3d^{10}\ 4s^2\ 4p^3$	121	947	2.0	Grey (metallic), Yellow, Black
Sb	[Kr] $4d^{10}\ 5s^2\ 5p^3$	141	834	1.9	Metallic (silver-grey), Yellow, Black
Bi	[Xe] $4f^{14}\ 5d^{10}\ 6s^2\ 6p^3$	148	703	1.9	White with pinkish tinge (only 1 form)

Table 15.1 — Physical and atomic properties of Group 15 elements.

KEY RESULT
IE-1 is substantially higher than Group 16 elements (e.g., $IE_1(N) = 1402 > IE_1(O) = 1314$ kJ/mol) due to the extra stability of the half-filled 2p^3 subshell in Nitrogen!

2. Anomalous Behaviour of Nitrogen

- **Small size & High Electronegativity:** Forms very strong **pπ–pπ triple bonds** (:N≡N: bond energy = 945.8 kJ/mol). Nitrogen exists as a chemically inert diatomic gas at RTP.
- **Single Bond Anomaly (N-N vs P-P):** N-N single bond energy (160 kJ/mol) is **much WEAKER** than P-P single bond energy (213 kJ/mol) due to intense inter-electronic repulsion between lone pairs in the compact 2p shell. Hence, **catenation in N is very limited** compared to P.
- **Absence of d-orbitals:** Nitrogen has a **maximum covalency of 4** (one 2s and three 2p orbitals). It CANNOT form pentahalides (NFS does not exist) or expand octet like P (PCl_5, PF_6^-).

3. Oxidation States & Inert Pair Effect

Common states: -3, +3, +5. Moving down the group: · **-3 state stability decreases:** Bi^{3-} is virtually unknown due to metallic electropositive nature. · **+5 state stability decreases:** Bi(+5) is unstable and a **powerful oxidising agent** (BiF_5 is known, but $BiCl_5$ does not exist; $NaBiO_3$ is a classic analytical oxidant for $Mn^{2+} \rightarrow MnO_4^-$). · **+3 state stability increases:** Bi(+3) is the most stable state of Bismuth due to the **inert pair effect**.

Dinitrogen (N_2) — Preparation, Inertness, Fixation & Chemical Reactions

Constitutes 78% of air by volume. Colorless, odorless, diamagnetic gas, condensation temp 77.2 K.

1. Laboratory & High-Purity Preparations

Method	Reaction Equation	Purity & Key Notes
Aqueous Nitrite Reaction	$NH_4Cl(aq) + NaNO_2(aq) \rightarrow \mathbf{N_2}\uparrow + NaCl + 2\ H_2O$	Standard lab prep; contains trace NO and HNO_3 impurities (purified by passing through aq. H_2SO_4 + $K_2Cr_2O_7$)
Thermal Decomposition of Dichromate	$(NH_4)_2Cr_2O_7 \xrightarrow{(\Delta)} \mathbf{N_2}\uparrow + Cr_2O_3$ (green solid) $+ 4\ H_2O$	Volcano reaction; exothermic decomposition of orange crystals
Thermal Azide Decomposition	$Ba(N_3)_2 \xrightarrow{(300°C)} Ba + \mathbf{3\ N_2}\uparrow$ or $2\ NaN_3 \rightarrow 2\ Na + \mathbf{3\ N_2}\uparrow$	**Ultra-pure N_2 (>99.99%)**; sodium azide used in automobile airbags!
Oxidation of Ammonia	$8\ NH_3 + 3\ Cl_2$ (excess NH_3) $\rightarrow \mathbf{N_2}\uparrow + 6\ NH_4Cl$ or $2\ NH_3 + 3\ CuO \rightarrow \mathbf{N_2}\uparrow + 3\ Cu + 3\ H_2O$	Yellow liquid NCl_3 formed if Cl_2 is in excess!

Table 15.2 — Synthetic routes to Dinitrogen.

KEY RESULT
Ultra-Pure Nitrogen is prepared exclusively by the **thermal decomposition of sodium or barium azide** NaN_3 / $Ba(N_3)_2$.

2. Chemical Reactivity of N2

Remarkably unreactive at room temperature because of the immense bond dissociation enthalpy of N≡N (945.8 kJ/mol). Becomes active only at elevated temperatures:

- **With Metals (Nitrides):** $6\ Li + N_2 \rightarrow 2\ Li_3N$ (ruby red ionic solid); $3\ Mg + N_2 \rightarrow Mg_3N_2$ (reacts with water to evolve NH_3: $Mg_3N_2 + 6\ H_2O \rightarrow 3\ Mg(OH)_2 + 2\ NH_3\uparrow$).
- **With Non-Metals:** · $N_2 + 3\ H_2 \rightleftharpoons 2\ NH_3$ (Haber process, 700 K, 200 atm, Fe/Mo catalyst, ΔH = -92.4 kJ/mol). · $N_2 + O_2 \rightleftharpoons 2\ NO$ (Electric arc, T > 2000 K, Birkeland-Eyde process, endothermic ΔH = +180 kJ/mol).
- **Nitrolim Formation:** $CaC_2 + N_2 \xrightarrow{(1373\ K)} \mathbf{CaCN_2 + C}$ ('Nitrolim' or Calcium Cyanamide). Used as an agricultural fertilizer; slowly hydrolyses in soil to urea and ammonia: $CaCN_2 + 3\ H_2O \rightarrow CaCO_3 + 2\ NH_3$.

Ammonia (NH3) — Haber Synthesis, Molecular Properties & Liquid NH3 Solvent

Trigonal pyramidal geometry (sp^3 hybridized, one lone pair, bond angle 107.8°, dipole moment 1.47 D). Associated in liquid and solid states via intermolecular hydrogen bonding.

1. Synthesis: The Haber-Bosch Process

KEY RESULT

$N_2(g) + 3\ H_2(g) \rightleftharpoons 2\ NH3(g)$ $\Delta H° = -92.4$ kJ/mol (Exothermic) Optimum Le Chatelier conditions: High pressure ($_{200\ atm}$), Moderate temperature ($_{700\ K}$), Catalyst: Iron oxide (FeO/Fe_2O_3) with promoter K_2O and Al $_2O_3$ (increases surface area and activity).

2. Chemical Reactions of NH3

- **Basic Nature & Salt Formation:** Weak base in water ($K_b = 1.8 \times 10^{-5}$). Forms ammonium salts with acids ($NH3 + HCl \rightarrow NH4Cl$ white dense fumes).
- **Precipitation of Metal Hydroxides:** $Fe^{3+} + 3\ NH4OH \rightarrow Fe(OH)_3\downarrow$ (reddish brown); $Zn^{2+} + 2\ NH4OH \rightarrow Zn(OH)_2\downarrow$ (white, dissolves in excess NH3 forming $[Zn(NH_3)_4]^{2+}$).
- **Complexation with Transition Metals (Deep Blue / Colour Changes):** · $Cu^{2+}(aq)$ (pale blue) + 4 NH3 → **$[Cu(NH_3)_4]^{2+}$** (tetraamminecopper(II), deep Prussian blue). · $AgCl\downarrow$ (white precipitate) + 2 NH3 → **$[Ag(NH_3)_2]^+$** + Cl^- (soluble diamminesilver(I)).
- **Nessler's Reagent Test:** Alkaline $K_2[HgI4] + NH3 \rightarrow$ **$[HgO{\cdot}Hg(NH_2)I]\downarrow$** (Iodide of Millon's base, brown precipitate). Extremely sensitive qualitative test for $NH4^+$.

3. Liquid Ammonia as a Non-Aqueous Ionising Solvent

Liquid NH3 (bp -33°C) autoionizes: $2\ NH3 \rightleftharpoons NH4^+$ (acid analogue) + $NH2^-$ (base analogue). Alkali metals dissolve in liquid ammonia giving spectacular solutions:

Concentration	Colour & Magnetic Property	Active Species & Behaviour
Dilute Solution (<3 M)	**Deep Blue**, strongly **paramagnetic**	**Ammoniated electrons $[e(NH3)_y]^-$**: Powerful reducing agent (Birch reduction)
Concentrated (>3 M)	**Bronze / Copper luster**, **diamagnetic**	Electrons pair up to form diamagnetic electron pairs; metallic electrical conductivity

Table IS.3 — Metal-ammonia solutions (NCERT & JD Lee core).

Nitrogen Hydrides & Derivatives — N_2H_4, NH2OH, HN3 & Redox Chemistry

Nitrogen forms a remarkable sequence of hydrides spanning oxidation states from -3 to -1/3.

1. Hydrazine (N2H4, H2N-NH2, Oxidation State = -2)

- **Raschig Synthesis:** 2 NH3 + NaOCl --(gelatin/glue catalyst)--→ **N_2H_4** + NaCl + H_2O. (Gelatin complexes with heavy metal ions like Cu^{2+} which would otherwise catalyse hydrazine decomposition!).
- **Structure:** Gauche conformation (dihedral angle ≈ 90° to minimise lone-pair repulsions, sp^3 N atoms).
- **Reactions & Uses:** Fuming colorless liquid, bifunctional weak base (pK_b1 = 6.07). Powerful **reducing agent**: $N_2H_4 + 2\ I_2 \rightarrow N_2\uparrow + 4\ HI$ (quantitative, no residue; hence used as rocket propellant fuel with N_2O_4).

2. Hydroxylamine (NH2OH, Oxidation State = -1)

Prepared by electrolytic reduction of nitric acid: $HNO3 + 6\ H^+ + 6\ e^- \rightarrow$ **NH2OH** $+ 2\ H_2O$. Acts as both oxidising and reducing agent: · Reducing action: $2\ NH2OH + 4\ Fe^{3+} \rightarrow N_2O + 4\ Fe^{2+} + 4\ H^+ + H_2O$. · Oxidising action: $NH2OH + 2\ Fe^{2+} + 2\ H^+ \rightarrow NH3 + 2\ Fe^{3+} + H_2O$. Forms **oximes** with aldehydes and ketones: $R_2C{=}O + NH2OH \rightarrow R_2C{=}N{-}OH + H_2O$.

3. Hydrazoic Acid & Azides (HN3, Oxidation State = -1/3)

Compound / Ion	Formal Oxidation State	Structure & Geometry	Safety / Stability
Hydrazoic acid HN3	-1/3	$H{-}N_1{=}N_2{=}N_3$ (linear N-N-N backbone, angle 171°)	Colorless, volatile liquid (bp 37°C), violently explosive!
Azide ion N_3^-	-1/3	$[:N{=}N{=}N:]^-$ (symmetric linear, D_∞h, N-N bond 116 pm)	$Pb(N_3)_2$ used in percussion caps and explosive detonators

Table 15.4 — Azide systems in inorganic chemistry.

KEY RESULT

Oxidation State Summary of Nitrogen Hydrides: · **NH3:** -3 | **N_2H_4:** -2 | **NH2OH:** -1 | **N_2:** 0 | **HN3:** -1/3

Oxides of Nitrogen — Structures, Oxidation States, Colors & Magnetic Properties

Nitrogen forms six classic oxides across oxidation states from +1 to +5. Several possess odd electrons.

1. Comprehensive Summary of Nitrogen Oxides

Oxide	Oxid. State	Prep Reaction	Colour & Physical State	Magnetic Nature
N_2O (Nitrous oxide)	+1	$NH_4NO_3 \xrightarrow{(\Delta,\ 250°C)} N_2O + 2\ H_2O$	Colorless gas (Laughing gas)	Diamagnetic (Neutral)
NO (Nitric oxide)	+2	$2\ NaNO_2 + 2\ FeSO_4 + 3\ H_2SO_4 \rightarrow Fe_2(SO_4)_3 + 2\ NaHSO_4 + 2\ H_2O + 2\ NO$	Colorless gas (Liquid is blue)	**Paramagnetic (gas)**, Diamag (liquid/solid dimer) (Neutral)
N_2O_3 (Sesquioxide)	+3	$2\ NO + N_2O_4 \xrightarrow{(250\ K)} 2\ N_2O_3$	**Deep Blue liquid** / solid	Diamagnetic (Acidic)
NO_2 (Dioxide)	+4	$2\ Pb(NO_3)_2 \xrightarrow{(\Delta,\ 673\ K)} 2\ PbO + 4\ NO_2 + O_2$	**Brown gas**	**Paramagnetic** (odd-electron molecule) (Acidic)
N_2O_4 (Tetroxide)	+4	$2\ NO_2 \xleftrightarrow{(cool\ /\ warm)} N_2O_4$	Colorless liquid / solid	Diamagnetic (Acidic)
N_2O_5 (Pentoxide)	+5	$4\ HNO_3 + P_4O_{10} \rightarrow 4\ HPO_3 + 2\ N_2O_5$	Colorless crystalline solid	Diamagnetic (Strongly Acidic)

Table IS.5 — Complete spectrum of Nitrogen Oxides (NCERT & JD Lee core).

KEY RESULT
Neutral Oxides: **N_2O and NO** are completely NEUTRAL to litmus! All other oxides (N_2O_3, NO_2, N_2O_4, N_2O_5) are acidic anhydrides.

2. Bonding & Structural Details

- **NO (Paramagnetic Monomer vs Diamagnetic Dimer):** Gas contains 11 valence electrons; unpaired electron resides in a π antibonding molecular orbital (bond order 2.5). In liquid and solid states, it dimerises to planar cis-O=N-N=O, pairing electrons (diamagnetic).
- **$NO_2 \rightleftharpoons N_2O_4$ Equilibrium:** NO_2 has a single odd electron on N (brown, paramagnetic, bent O-N-O angle 134°). On cooling, two NO_2 molecules dimerise to form planar N_2O_4 with an unusually long N-N bond (175 pm vs normal 145 pm) due to repulsion between bonding pairs.
- **Solid N_2O_5 Ionic Lattice:** In the gas phase, N_2O_5 is molecular (O_2N-O-NO_2). In the **solid crystalline state**, it exists as an ionic salt: **$[NO_2]^+$ $[NO_3]^-$ (Nitronium nitrate)**! Linear $[NO_2]^+$ cation + planar trigonal $[NO_3]^-$ anion.

TRAP
TRAP: Liquid N_2O_4 undergoes auto-ionization: $N_2O_4 \rightleftharpoons NO^+$ (nitrosyl cation) + NO_3^- (nitrate anion). It acts as a non-aqueous solvent where NOCl acts as an acid and $AgNO_3$ acts as a base!

Oxoacids of Nitrogen — Nitrous Acid, Nitric Acid & The Brown Ring Test Mechanism

Primary oxoacids: Nitrous acid HNO2 (+3, unstable, weak acid) and Nitric acid HNO3 (+5, strong acid & oxidising agent).

1. Nitrous Acid (HNO2) & Disproportionation

Prepared freshly in cold: $NaNO_2 + HCl \xrightarrow{(0\text{-}5°C)} NaCl + HNO_2$. On warming, it **disproportionates**: $3\,HNO_2 \xrightarrow{(warm)} \mathbf{HNO_3\ (+5) + 2\,NO\ (+2) + H_2O}$. Acts as both oxidising agent (oxidises I^- to I_2) and reducing agent (decolourises $KMnO_4$).

2. Nitric Acid (HNO3) Manufacture: Ostwald Process

- **Step 1 (Catalytic Oxidation):** $4\,NH_3(g) + 5\,O_2(g) \xrightarrow{(Pt\text{-}Rh\ gauze,\ 1150\ K,\ 9\ bar)} \mathbf{4\,NO(g)} + 6\,H_2O(g)$ (exothermic).
- **Step 2 (Dimerization / Oxidation):** $2\,NO(g) + O_2(g) \rightleftharpoons \mathbf{2\,NO_2(g)}$.
- **Step 3 (Absorption in Water):** $3\,NO_2(g) + H_2O(l) \rightarrow \mathbf{2\,HNO_3(aq) + NO(g)}$ (NO is recycled back to Step 2).

3. Oxidising Action of HNO3 on Metals

Metal	Concentration of HNO3	Oxidation Product & Nitrogen Gas Evolved
Copper (Cu)	Dilute HNO3 (~20%)	$3\,Cu + 8\,HNO_3 \rightarrow 3\,Cu(NO_3)_2 + 2\,H_2O + \mathbf{2\,NO}\uparrow$ (Nitric oxide)
Copper (Cu)	Concentrated HNO3 (~70%)	$Cu + 4\,HNO_3 \rightarrow Cu(NO_3)_2 + 2\,H_2O + \mathbf{2\,NO_2}\uparrow$ (Brown gas)
Zinc (Zn)	Dilute HNO3 (~20%)	$4\,Zn + 10\,HNO_3 \rightarrow 4\,Zn(NO_3)_2 + 5\,H_2O + \mathbf{N_2O}\uparrow$ (Nitrous oxide)
Zinc (Zn)	Concentrated HNO3 (~70%)	$Zn + 4\,HNO_3 \rightarrow Zn(NO_3)_2 + 2\,H_2O + \mathbf{2\,NO_2}\uparrow$ (Brown gas)
Zinc (Zn)	Very Dilute HNO3 (~6%)	$4\,Zn + 10\,HNO_3 \rightarrow 4\,Zn(NO_3)_2 + 3\,H_2O + \mathbf{NH_4NO_3}$ (Ammonium nitrate)

Table 15.6 — Reaction of metals with dilute vs concentrated Nitric Acid (JEE Classic).

4. Mechanism of the Brown Ring Test for Nitrate (NO_3^-)

Aqueous nitrate + freshly prepared $FeSO_4$ + conc. H_2SO_4 carefully poured down the side of the test tube: 1. Reduction: $NO_3^- + 3\,Fe^{2+} + 4\,H^+ \rightarrow 3\,Fe^{3+} + \mathbf{NO} + 2\,H_2O$. 2. Coordination & Complex Formation: $[Fe(H_2O)_6]^{2+} + NO \rightarrow \mathbf{[Fe(H_2O)_5(NO)]^{2+}}$ (Brown Ring) + H_2O.

KEY RESULT

Oxidation State of Fe in Brown Ring Complex $[Fe(H_2O)_5(NO)]^{2+}$: One electron is transferred from NO to Fe^{2+}, creating **Fe(+1) (d^7, $t_{2g}^5\,e_g^{2)**}$ and $^{**}NO^+$ (nitrosonium ion)**! Magnetic Moment: $\mu = \sqrt{15} \approx$ **3.87 BM (3 unpaired electrons)**. This is a critical JEE Advanced question.

Phosphorus Allotropes & Phosphine — White, Red, Black Allotropy & PH3 Smoke Rings

Phosphorus exists in three major allotropic forms: White (P_4), Red (polymeric), and Black (layered).

1. Comparative Allotropy of Phosphorus

Property	White Phosphorus	Red Phosphorus	Black Phosphorus (α and β)
Molecular Structure	Discrete P_4 tetrahedra (angle 60°)	Complex 3D polymer of linked P_4 rings	Layered puckered hexagonal sheets (flaky)
Physical State	Translucent waxy solid (soft, cuts with knife)	Reddish-violet powdery solid (odorless)	Black lustrous flaky solid (graphite-like)
Toxicity & Chemiluminescence	**Highly toxic**, glows green in dark (chemiluminescence)	**Non-toxic**, does NOT glow in dark	Non-toxic, completely non-reactive
Reactivity & Ignition	Spontaneously catches fire in air at 30°C (kept under water!)	Stable in air, ignites at 260°C	Least reactive, highly inert
Solubility	Insoluble in H_2O, soluble in CS2 and benzene	Insoluble in both H_2O and CS2	Insoluble in all ordinary solvents

Table 15.7 — Comparison of Phosphorus allotropes (NCERT core).

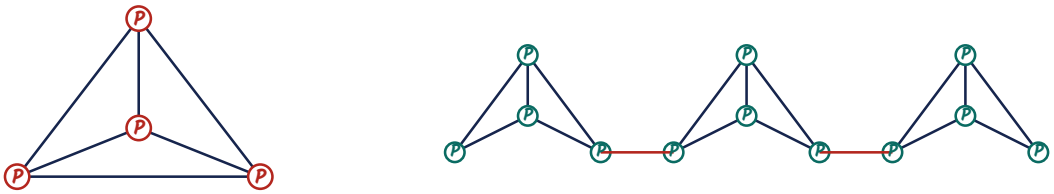


Fig. 15.1 — White P_4 discrete tetrahedron vs Red Phosphorus polymeric chain.

2. Phosphine (PH_3) — Synthesis & Reactions

- **Preparation (Disproportionation of White P_4):** $P_4 + 3\ NaOH + 3\ H_2O$ --(inert CO2 atm)--→ **$PH_3\uparrow$ + 3 NaH2PO2** (Sodium hypophosphite, +1 state). (Pure PH3 is non-flammable; it catches fire spontaneously due to traces of **P_2H_4 (diphosphane)** impurity!).
- **Holmes Smoke Signals:** Containers of CaC2 + Ca_3P_2 pierced and thrown into sea. Water reacts: $Ca_3P_2 + 6\ H_2O \rightarrow 3\ Ca(OH)_2 + 2\ PH_3\uparrow$ (contains P_2H_4 → ignites → produces vortex smoke rings of P_4O_{10}). $CaC_2 + 2\ H_2O \rightarrow Ca(OH)_2 + C_2H_2\uparrow$ (acetylene burns giving bright illuminating flame).
- **Basicity:** PH3 is a very weak base (much weaker than NH3, no H-bonding, Drago's rule: unhybridized pure p-orbital lone pair). Only forms salts with strong acids: $PH_3 + HI \rightarrow$ **PH4I** (Phosphonium iodide).

Phosphorus Halides — PCl3, PCl5, Axial/Equatorial Bonds & Solid-State Ionization

Phosphorus forms trihalides PX3 (sp^3 pyramidal) and pentahalides PX5 (sp^3d trigonal bipyramidal).

1. Phosphorus Trichloride (PCl3)

- **Synthesis:** $P_4 + 6\ Cl_2$ (dry, limited) → 4 PCl3 (or: $P_4 + 8\ SOCl2 \rightarrow 4\ PCl3 + 4\ SO2 + 2\ S_2Cl_2$).
- **Structure:** Pyramidal (sp^3 P, one lone pair, P-Cl bond length 204 pm).
- **Hydrolysis:** $PCl3 + 3\ H_2O \rightarrow$ **H_3PO_3** + 3 HCl (violent, fumes copiously in moist air).
- **Reactions with Organic OH:** $3\ C_2H_5OH + PCl3 \rightarrow 3\ C_2H_5Cl + H_3PO_3$; $3\ CH3COOH + PCl3 \rightarrow 3\ CH3COCl + H_3PO_3$.

2. Phosphorus Pentachloride (PCl5) — Gas vs Solid State

State of Matter	Chemical Structure & Species Present	Bond Lengths & Geometry
Gas Phase & Liquid	Discrete molecular PCl5	Trigonal bipyramidal (sp^3d): **3 equatorial bonds (202 pm) + 2 axial bonds (240 pm)**. Axial bonds are longer and weaker due to 3 × 90° repulsions!
Solid Crystalline State	**Ionic Solid:** $[PCl4]^+$ $[PCl6]^-$	$[PCl4]^+$ is **tetrahedral (sp^3)**; $[PCl6]^-$ is **octahedral (sp^3d^2)**!
PBr5 Solid State	**Ionic Solid:** $[PBr4]^+$ $[Br]^-$	$[PBr4]^+$ tetrahedral cation + Br^- anion (P is too small to coordinate 6 bulky Br atoms!)

Table 15.8 — Gas-phase vs Solid-state structures of Phosphorus pentahalides.

KEY RESULT

Why are axial bonds longer than equatorial bonds in PCl5? Equatorial P-Cl bonds experience repulsion from only two adjacent bonds at 90°. Each axial P-Cl bond experiences repulsion from **three equatorial bonds at 90°**, making them longer and weaker!

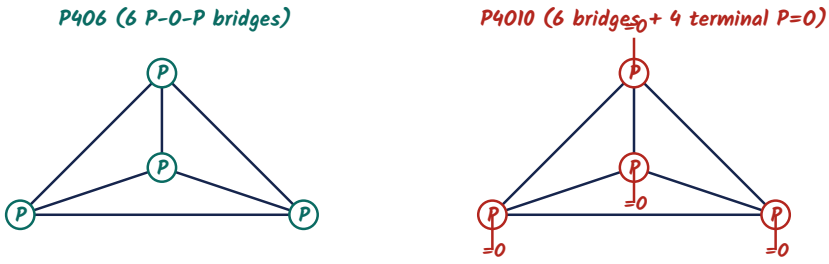
3. Hydrolysis of PCl5

Stepwise hydrolysis: • Step 1 (Limited water): $PCl5 + H_2O \rightarrow$ **POCl3 (Phosphorus oxychloride)** + 2 HCl. • Step 2 (Excess water): $POCl3 + 3\ H_2O \rightarrow$ **H_3PO_4 (Orthophosphoric acid)** + 3 HCl. Net: **$PCl5 + 4\ H_2O \rightarrow H_3PO_4 + 5$ HCl**.

Oxides of Phosphorus — P_4O_6, P_4O_{10}, Adamantane Cages & Dehydrating Power

Prepared by burning white phosphorus in air/oxygen: limited air gives P_4O_6, excess air gives P_4O_{10}.

1. Structure & Bonding of P4O6 and P4O10



Each P in P4O10 is sp3 tetrahedral; 16 total P-O bonds (6 bridging P-O-P + 4 P=O)

Fig. 15.2 — Adamantane cage architecture of P_4O_6 and P_4O_{10}.

Feature	Phosphorus Trioxide (P_4O_6)	Phosphorus Pentoxide (P_4O_{10})
Combustion Condition	$P_4 + 3\ O_2$ (limited) → P_4O_6	$P_4 + 5\ O_2$ (excess) → P_4O_{10}
Formal Oxidation State	+3	+5
Number of P-O-P Bridges	6 bridging P-O-P bonds	6 bridging P-O-P bonds
Terminal P=O Bonds	0 (each P retains one lone pair)	4 terminal P=O bonds (dπ-pπ backbonding)
Total Number of P-O Bonds	12 P-O single bonds	**16 P-O bonds** (12 bridging + 4 terminal)
Hybridization of P	sp^3 (pyramidal at each P)	sp^3 (tetrahedral at each P)

Table 15.9 — Structural parameters of P_4O_6 vs P_4O_{10} cages.

KEY RESULT

Counting Bonds in P_4O_{10} (Guaranteed JEE Question): · Bridging P-O-P linkages = **6** · Terminal P=O double bonds = **4** · Total P-O single bond vectors in cage = **12** (plus 4 terminal = 16 total)

2. Hydrolysis Pathways & Acid Anhydrides

- **Hydrolysis of P_4O_6:** $P_4O_6 + 6\ H_2O$ (cold) → **$4\ H_3PO_3$ (Phosphorous acid)**. (With hot water, disproportionates: $P_4O_6 + 6\ H_2O \rightarrow 3\ H_3PO_4 + PH_3\uparrow$).
- **Hydrolysis of P_4O_{10}:** Stepwise hydration: P_4O_{10} --(+2 H_2O)--→ 4 HPO3 (Metaphosphoric acid) --(+2 H_2O)--→ 2 $H_4P_2O_7$ (Pyrophosphoric acid) --(+2 H_2O)--→ **$4\ H_3PO_4$ (Orthophosphoric acid)**.
- **Powerful Dehydrating Action of P_4O_{10}:** It strips water from acids to generate anhydrides: · 2 HNO3 + P_4O_{10} → **N_2O_5** + 2 HPO3 · 2 H_2SO_4 + P_4O_{10} → **2 SO3** + 4 HPO3 · 2 CH3CONH2 + P_4O_{10} → **2 CH3CN (Acetonitrile)** + 4 HPO3 · (COOH)2 + P_4O_{10} → **CO + CO2** + 2 HPO3

Oxoacids of Phosphorus — Basicity, P-H Bonds & Reducing Power

Phosphorus forms a famous series of oxoacids characterized by tetrahedral sp^3 P atoms with at least one P=O bond and one P-OH bond. Basicity equals the number of P-OH bonds!

1. Comprehensive Oxoacids Table

Oxoacid	Formula	State	Basicity	No. of P-H	Reducing Character
Hypophosphorous acid	H_3PO_2	+1	**1 (Monobasic)**	**2 P-H bonds**	Very Strong (reduces AgNO3 → Ag↓)
Orthophosphorous acid	H_3PO_3	+3	**2 (Dibasic)**	**1 P-H bond**	Strong (reduces CuSO4 → Cu↓)
Orthophosphoric acid	H_3PO_4	+5	**3 (Tribasic)**	**0 P-H bonds**	Non-reducing
Pyrophosphoric acid	$H_4P_2O_7$	+5	**4 (Tetrabasic)**	0 P-H bonds	Non-reducing (P-O-P bridge)
Hypophosphoric acid	$H_4P_2O_6$	+4	**4 (Tetrabasic)**	0 P-H bonds	Non-reducing (**P-P direct bond!**)
Cyclotrimetaphosphoric	(HPO3)3	+5	**3 (Tribasic)**	0 P-H bonds	Non-reducing (3 P-O-P bridges)

Table 15.10 — Structural formulas and properties of Phosphorus oxoacids.

KEY RESULT

Basicity vs Reducing Rule: · **Basicity = Number of P-OH bonds** (only H attached to oxygen is ionisable as H^+). · **Reducing Power = Number of P-H bonds** (hydrogenic H attached directly to P is reducing!).

2. Diagnostic Disproportionation of Intermediate Acids

Acids containing phosphorus in intermediate oxidation states (+1 to +4) **disproportionate** on heating: · 4 H_3PO_3 (+3) --(Δ, 200°C)--→ **3 H_3PO_4 (+5) + PH3 (-3)** · 2 H_3PO_2 (+1) --(Δ, 160°C)--→ **H_3PO_4 (+5) + PH3 (-3)**

3. Quantitative Reaction with Silver Nitrate

Hypophosphorous acid H_3PO_2 has two P-H bonds and acts as a powerful analytical reducing agent: 4 AgNO3 + 2 H_2O + H_3PO_2 → **4 Ag↓ (black metallic mirror) + H_3PO_4 + 4 HNO3**.

Group 16: Chalcogens ($ns^2\ np^4$) — Ore Formers & Periodic Trends

Contains O, S (non-metals), Se, Te (metalloids), Po (radioactive α-emitter metal). The electronic configuration has two unpaired p-electrons. Key themes: anomalous oxygen properties, catenation in sulphur, and pπ-dπ vs pπ-pπ multiple bonding.

1. Periodic Trends of Group 16

Element	Config.	Covalent Radius (pm)	IE-1 (kJ/mol)	ΔH_eg (kJ/mol)	Electronegativity
O	[He] $2s^2\ 2p4$	66	1314	-141	3.5 (2nd highest after F)
S	[Ne] $3s^2\ 3p4$	104	1000	**-200 (Most negative!)**	2.5
Se	[Ar] 3d¹⁰ 4s2 4p^4	117	941	-195	2.4
Te	[Kr] 4d¹⁰ 5s2 5p^4	137	869	-190	2.1
Po	[Xe] 4f¹⁴ 5d10 $6s^2\ 6p4$	146	813	-174	2.0

Table 16.1 — Physical and atomic parameters of Group 16 elements.

KEY RESULT
Electron Gain Enthalpy Anomaly: **S (-200) > Se (-195) > Te (-190) > Po (-174) > O (-141 kJ/mol)**. Oxygen has the LEAST negative electron gain enthalpy in the entire group because of its very compact 2p subshell, causing severe inter-electronic repulsions for incoming electrons!

2. Catenation & Allotropy

Sulphur shows maximum catenation in Group 16 because the S-S single bond (213 kJ/mol) is far stronger than the O-O single bond (142 kJ/mol). Polysulphide ions like S_n^{2-} (e.g. S_2^{2-}, S_5^{2-}) and polythionates exist stably in aqueous solution.

3. Multiple Bonding: pπ-pπ vs pπ-dπ

- **Oxygen:** Forms strong **pπ-pπ double bonds** (O=O, C=O, N=O) due to compact 2p orbitals. Hence, oxygen exists as a diatomic gas O_2 with weak van der Waals forces.
- **Sulphur & Heavier Chalcogens:** Cannot form stable pπ-pπ bonds; instead, they form single bonds with extensive catenation, leading to polyatomic crown molecules like **S_8** (solid at RTP).
- **d-Orbital Participation:** Sulphur expands its octet using vacant 3d orbitals, forming hexa-coordinate molecules like **SF6** and utilizing **pπ-dπ back-bonding** in SO2 and SO3.

Dioxygen & Ozone — Paramagnetism of O_2 & Strong Oxidising Action of O_3

Oxygen exists as triplet ground-state O_2 (paramagnetic) and triatomic allotrope Ozone O_3 (diamagnetic).

1. Dioxygen (O2) — Molecular Orbital Configuration

Valence electron count = 12 electrons. MO configuration: $\sigma(2s)^2\ \sigma^*(2s)^2\ \sigma(2p_z)^2\ \pi(2p_x)^2\ \pi(2p_y)^2\ {}^{**}\pi^*(2p_x)^1\ \pi(2p_y)^{\wedge}1$. **Bond order = (8 - 4)/2 = 2. It possesses** two unpaired electrons **in degenerate** π **antibonding orbitals, explaining its experimental** paramagnetism.

2. Ozone (O3) — Synthesis & Angular Resonance Structure

- **Synthesis (Silent Electric Discharge):** $3\ O_2 \rightleftharpoons 2\ O_3$ ($\Delta H° = +142$ kJ/mol, endothermic). A silent electric discharge is used to prevent local heating which would decompose O_3 back to O_2.
- **Structure:** V-shaped angular geometry (C_{2v} symmetry), $sp^{\wedge}2$ hybridized central O atom, bond angle **117°**, O-O bond length **128 pm** (intermediate between single bond 148 pm and double bond 121 pm due to resonance).
- **Thermodynamics of Ozone:** Thermodynamically unstable: $2\ O_3 \rightarrow 3\ O_2$ (ΔH 0, so **ΔG is immensely negative**). High concentrations can decompose explosively!

3. Powerful Oxidising Action & Qualitative Tests for Ozone

Substance Oxidised	Reaction Equation	Observation / Diagnostic Test
Lead Sulphide (black)	PbS (black) + $4\ O_3 \rightarrow$ **$PbSO_4$ (white)** + $4\ O_2$	Decolourisation of black lead sulphide paper
Potassium Iodide (KI)	$2\ KI + H_2O + O_3 \rightarrow$ **$2\ KOH + I_2 + O_2$**	Turns starch paper blue (Quantitative iodometric estimation: titrate I_2 with $Na_2S_2O_3$)
Mercury Metal (Hg)	**$2\ Hg + O_3 \rightarrow Hg_2O + O_2$**	**Tailing of Mercury:** Hg loses its meniscus and sticks to glass surface due to Hg_2O formation. Restored by shaking with water!
Potassium Manganate	$2\ K_2MnO_4$ (green) + $H_2O + O_3 \rightarrow$ **$2\ KMnO_4$ (purple)** + $2\ KOH + O_2$	Oxidation of green manganate to purple permanganate

Table 16.2 — Diagnostic oxidising reactions of Ozone.

KEY RESULT

Tailing of Mercury Test: $2\ Hg + O_3 \rightarrow Hg_2O + O_2$. The loss of meniscus is a unique and specific test for Ozone in JEE.

Allotropes of Sulphur — Rhombic (α), Monoclinic (β), Chair S_6 & Paramagnetic S_2

Sulphur exhibits more allotropes than almost any other element. The most important crystalline forms are Rhombic (α) and Monoclinic (β).

1. Rhombic (α-Sulphur) vs Monoclinic (β-Sulphur)

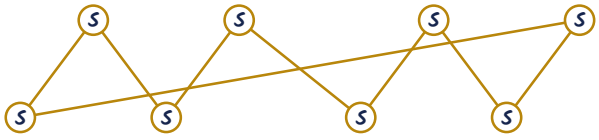
Property	Rhombic Sulphur (α)	Monoclinic Sulphur (β)
Crystal System	Orthorhombic (octahedral crystals)	Monoclinic (needle-shaped crystals)
Colour & Density	Bright yellow, density 2.06 g/cm^3	Amber yellow, density 1.96 g/cm^3
Melting Point	385.8 K (112.8°C)	393 K (119.2°C)
Stability Range	**Stable below 369 K (96°C)**	**Stable above 369 K (96°C)**
Solubility	Insoluble in H_2O, soluble in CS2 (~30 g/100g)	Insoluble in H_2O, soluble in CS2

Table 16.3 — Comparison of Rhombic and Monoclinic Sulphur.

KEY RESULT

Transition Temperature (369 K / 95.6°C): Below 369 K, α-sulphur is stable. Above 369 K, β-sulphur is stable. At exactly 369 K, both allotropes coexist in dynamic equilibrium (Enantiotropic allotropy).

2. Molecular Geometry: Crown S8 & Chair S6



S8 Crown Ring (D4d symmetry)
S–S bond length = 204 pm, S–S–S bond angle = 107 deg

Fig. 16.1 — Puckered S_8 crown ring (D_{4d}) with bond parameters.

- **Puckered S_8 Crown:** Both α and β sulphur consist of puckered S_8 crown rings (D_{4d} symmetry). S-S bond distance = 204 pm, S-S-S angle = 107°.
- **Engels Sulphur (S_6):** Prepared by reacting $Na_2S_2O_3$ with conc. HCl. Contains a puckered **chair conformation** (like cyclohexane) with S-S-S angle = 102.2° and S-S distance = 205.7 pm.
- **High Temperature S_2 Molecule (Paramagnetism):** At elevated temperatures (> 1000 K), sulphur vapor exists predominantly as diatomic **S_2** molecules. Just like O_2, S_2 has **two unpaired electrons** in its π antibonding molecular orbitals and is **paramagnetic**!

3. Plastic (Amorphous) Sulphur

Formed by pouring boiling liquid sulphur (at 440 K, dark brown viscous liquid with broken S_8 chains) in a thin stream into cold water. Yellow-brown rubbery elastic mass containing tangled helical polymer chains (S_μ). It is insoluble in CS2; slowly reverts to rhombic α-sulphur on standing.

Group 16 Hydrides (H_2E) — Volatility, Acidity, Bond Angles & Drago's Rule

All chalcogens form volatile hydrides H_2E (E = O, S, Se, Te, Po). Properties show sharp trends.

1. Periodic Trends in Hydrides

Property	H_2O	H_2S	H_2Se	H_2Te	Reason / Principle
Boiling Point (K)	**373**	213	232	269	H_2O is anomalously high due to extensive 3D intermolecular H-bonding; then increases with molar mass (dispersion forces): **$H_2S < H_2Se < H_2Te < H_2O$**
Bond Angle (°)	**104.5°**	92.1°	91.0°	90.0°	**Drago's Rule:** In H_2S, H_2Se, H_2Te, no sp^3 hybridization occurs; bonds formed by almost pure p orbitals (approx 90°)!
Thermal Stability	**High**	Moderate	Low	Very Poor	E-H bond dissociation energy decreases down the group as orbital overlap becomes diffuse: **$H_2O > H_2S > H_2Se > H_2Te$**
Reducing Character	Nil	Mild	Strong	**Strongest**	Follows inverse of thermal stability: **$H_2O < H_2S < H_2Se < H_2Te$**
Acidic Strength (K_a)	10^{-14}	10^{-7}	10^{-4}	**10^{-3}**	E-H bond length increases and bond energy decreases: **$H_2O < H_2S < H_2Se < H_2Te$**

Table 16.4 — Systematic trends in Group 16 hydrides.

KEY RESULT

Critical Order Sequences: · **Boiling Point:** H_2S H_2S > H_2Se > H_2Te · **Reducing Power:** H_2O < H_2S < H_2Se < **H_2Te**

2. Hydrogen Sulphide (H2S) in Qualitative Analysis

Rotten egg smell, poisonous gas. Precipitates metal cations as insoluble sulphides based on K_{sp}: · **Group II Cations (Acidic medium, dil. HCl):** Low $[S^{2-}]$ due to common ion effect of H^+; precipitates low K_{sp} sulphides (CuS, PbS, Bi_2S_3, CdS, HgS). · **Group IV Cations (Basic medium, NH4OH + NH4Cl):** High $[S^{2-}]$ due to removal of H^+ by OH^-; precipitates higher K_{sp} sulphides (ZnS, NiS, CoS, MnS).

Oxides of Sulphur — SO2, SO3 Polymerism & Temporary Bleaching Action

Sulphur forms two commercially crucial oxides: Sulphur dioxide SO2 and Sulphur trioxide SO3.

1. Sulphur Dioxide (SO2) — Structure & Bleaching

- **Synthesis:** $S + O_2 \rightarrow SO2$ (industrial) or $Cu + 2\ H_2SO_4$ (conc.) $\xrightarrow{(\Delta)} CuSO4 + SO2\uparrow + 2\ H_2O$ (lab).
- **Structure:** V-shaped angular (sp^2 S, one lone pair, O-S-O angle 119.5°, bond length 143 pm equalized by resonance involving pπ-pπ and pπ-dπ bonding).
- **Bleaching Action (Temporary by Reduction):** In the presence of moisture, SO2 acts as a reducing bleaching agent: $SO2 + 2\ H_2O \rightarrow H_2SO_4 +$ **2 [H] (Nascent Hydrogen)** Coloured organic matter + [H] → **Colourless product**. (On exposure to atmospheric oxygen, the bleached matter is re-oxidised back to coloured state. Contrast with Cl_2 bleaching, which is permanent by oxidation!).

Reagent	Reaction with SO2	Diagnostic Observation
Acidified $K_2Cr_2O_7$ (orange)	$K_2Cr_2O_7 + H_2SO_4 + 3\ SO2 \rightarrow$ **$Cr_2(SO_4)_3$ (green)** $+ K_2SO_4 + H_2O$	Orange solution turns emerald green (Test for SO2!)
Acidified KMnO4 (purple)	$2\ KMnO4 + 2\ H_2O + 5\ SO2 \rightarrow K_2SO_4 +$ **2 MnSO4 (colorless)** $+ 2\ H_2SO_4$	Pink/purple permanganate decolourises completely
Ferric Chloride (FeCl3)	$2\ FeCl3 + SO2 + 2\ H_2O \rightarrow$ **2 FeCl2 (pale green)** $+ H_2SO_4 + 2\ HCl$	Yellow ferric solution reduced to green ferrous
Iodine solution (I_2)	$I_2 + SO2 + 2\ H_2O \rightarrow$ **2 HI (colorless)** $+ H_2SO_4$	Brown iodine solution decolourises

Table 16.5 — Characteristic redox reactions of Sulphur Dioxide.

2. Sulphur Trioxide (SO3) & Solid Allotropes

In the gas phase, SO3 is planar trigonal (sp^2, D_{3h} symmetry, three equivalent S=O bonds). In the solid state, it undergoes remarkable polymerisation: · **γ-SO3 (Trimer):** Cyclic trimer **$(SO3)_3$** or S_3O_9 with a puckered 6-membered ring of alternating S and O atoms. · **β-SO3 & α-SO3 (Linear Polymers):** Infinite helical chains of tetrahedral SO4 units sharing corners, resembling asbestos fibers (mp 32°C and 62°C).

Oxoacids of Sulphur (Part I) — Contact Process & H_2SO_4 Dehydrating Action

Sulphuric acid H_2SO_4 ('King of Chemicals') is manufactured worldwide via the Contact Process.

1. Contact Process for H2SO4 Manufacture

- **Step 1:** Burning sulphur or iron pyrites in air: $S + O_2 \rightarrow SO2$ (or $4\ FeS2 + 11\ O_2 \rightarrow 2\ Fe_2O_3 + 8\ SO2$).
- **Step 2 (The Catalytic Key Step):** Catalytic oxidation of SO2 to SO3: $\mathbf{2\ SO2(g) + O_2(g) \leftarrow(V_2O_5,\ 720\ K,\ 2\ bar)\rightarrow 2\ SO3(g)}$ $\Delta H° = -196.6$ kJ/mol (Exothermic). (Optimal conditions: excess O_2, 2 bar pressure, 720 K with Vanadium pentoxide V_2O_5 catalyst).
- **Step 3:** Absorption of SO3 in 98% H_2SO_4 to form **Oleum (Pyrosulphuric acid $H_2S_2O_7$)**: $SO3 + H_2SO_4 \rightarrow \mathbf{H_2S_2O_7}$. (Direct absorption in water is avoided because it forms a dense, unmanageable acid mist!).
- **Step 4:** Dilution of oleum with calculated quantity of water: $H_2S_2O_7 + H_2O \rightarrow \mathbf{2\ H_2SO_4}$.

2. Chemical Properties & Dehydrating Action of H2SO4

Reaction Category	Substrate	Reaction Equation & Observation
Dehydrating Action	Cane Sugar (Sucrose $C_{12}H_{22}O_{11}$)	$C_{12}H_{22}O_{11}$ --(conc. H_2SO_4)--→ **12 C (black spongy char) + 11 H_2O** (Charring of sugar)
Dehydrating Action	Ethanol (C_2H_5OH)	C_2H_5OH --(conc. H_2SO_4, 443 K)--→ **C_2H_4 (Ethene)** + H_2O
Dehydrating Action	Formic Acid (HCOOH)	HCOOH --(conc. H_2SO_4)--→ **CO**↑ + H_2O
Dehydrating Action	Oxalic Acid ((COOH)2)	(COOH)2 --(conc. H_2SO_4)--→ **CO**↑ **+ CO2**↑ + H_2O
Oxidising Action	Copper (Cu)	$Cu + 2\ H_2SO_4$ (conc.) --(Δ)--→ $CuSO4 + \mathbf{SO2}↑ + 2\ H_2O$
Oxidising Action	Carbon (C) & Sulphur (S)	$C + 2\ H_2SO_4 \rightarrow \mathbf{CO2 + 2\ SO2} + 2\ H_2O$; $S + 2\ H_2SO_4 \rightarrow \mathbf{3\ SO2} + 2\ H_2O$

Table 16.6 — Diagnostic dehydrating and oxidising reactions of Sulphuric Acid.

KEY RESULT

Dilution Rule: Always add **acid slowly to water with constant stirring**, NEVER water to concentrated acid (huge exothermic hydration causes dangerous boiling and acid splattering!).

Oxoacids of Sulphur (Part II) — Peroxoacids, Thionic Series & Thiosulphate

Peroxoacids contain the -O-O- peroxo linkage (formal oxidation state of peroxo oxygens = -1).

1. Peroxomonosulphuric Acid (Caro's Acid, H2SO5)

Prepared by reacting chlorosulphonic acid with hydrogen peroxide: $ClSO_3H + H_2O_2 \rightarrow \mathbf{H_2SO_5} + HCl$. Structural Formula: **HO-SO2-O-O-H**. Formal Oxidation State Calculation: 2(+1) + S + 3(-2) + 2(-1) = 0 → **S = +6** (NOT +8!). Contains one peroxo (-O-O-) bond, two S=O bonds, and one normal S-OH bond.

2. Peroxodisulphuric Acid (Marshall's Acid, H2S2O8)

Prepared by anodic oxidation of cold 50% H_2SO_4 at high current density: $2\ HSO_4^- \rightarrow \mathbf{H_2S_2O_8} + 2e^-$. Structural Formula: **HO-SO2-O-O-SO2-OH**. Oxidation State of both Sulphur atoms = **+6** (contains one bridging -O-O- peroxide bond). Hydrolysis: $H_2S_2O_8 + H_2O \rightarrow H_2SO_5 + H_2SO_4$; $H_2SO_5 + H_2O \rightarrow H_2SO_4 + \mathbf{H_2O_2}$ (industrial route to H_2O_2).

Oxoacid	Formula	S Oxidation State	Key Structural Linkage	Basicity
Sulphurous acid	H_2SO_3	+4	One lone pair on S, one S=O, two S-OH	2 (Dibasic)
Dithionous acid	$H_2S_2O_4$	+3	Direct **S-S bond**, two S=O, two S-OH	2 (Dibasic)
Dithionic acid	$H_2S_2O_6$	+5	Direct **S-S bond**, four S=O, two S-OH	2 (Dibasic)
Polythionic acid	$H_2S_nO_6$	Average	Polysulphide chain **-S_(n-2)-** flanked by -SO3H	2 (Dibasic)
Pyrosulphuric (Oleum)	$H_2S_2O_7$	+6	**S-O-S bridge**, four S=O, two S-OH	2 (Dibasic)
Caro's acid	H_2SO_5	+6	**One -O-O- bridge**, two S=O, one S-OH	1 (Monobasic!)
Marshall's acid	$H_2S_2O_8$	+6	**One -O-O- bridge**, four S=O, two S-OH	2 (Dibasic)

Table 16.7 — Master classification of Sulphur oxoacids.

3. Sodium Thiosulphate (Na2S2O3·5H2O, 'Hypo')

- **Synthesis:** $Na_2SO_3 + S$ --(boil with water)--> $Na_2S_2O_3$. Structure: Tetrahedral with one O replaced by S; central S is +5, terminal S is -1 (Average = +2).
- **Photography (Fixer):** $AgBr + 2\ Na_2S_2O_3 \rightarrow \mathbf{Na_3[Ag(S_2O_3)_2]}$ **(soluble complex)** $+ NaBr$.
- **Iodometry:** $2\ Na_2S_2O_3 + I_2 \rightarrow \mathbf{Na_2S_4O_6}$ **(Sodium tetrathionate)** $+ 2\ NaI$.
- **With Strong Oxidant (Cl_2):** $Na_2S_2O_3 + 4\ Cl_2 + 5\ H_2O \rightarrow \mathbf{2\ NaHSO_4 + 8\ HCl} + \mathbf{H_2SO_4}$ (acts as 'Antichlor').

Sulphur Halides & Oxohalides — SF4 (See-Saw), SF6 (Inert Octahedron) & Thionyl Chloride

Sulphur forms hexafluoride SF6, tetrahalides SF4, SCl4, and dihalides SCl2, S_2Cl_2.

1. Sulphur Hexafluoride (SF6) — Kinetic Inertness

- **Structure:** Regular Octahedral ($sp^{3d}2$, O_h symmetry, all S-F bonds = 156 pm).
- **Extraordinary Chemical Inertness:** SF6 is non-toxic, odorless, completely inert to water, boiling molten KOH, and acids. Does not hydrolyse even at 500°C!
- **Reason for Inertness:** S atom is coordinatively saturated and sterically shielded by six compact Fluorine atoms. Water molecules cannot approach the central Sulphur atom, and Sulphur cannot expand its coordination sphere further.
- **Applications:** High dielectric strength gas used as an electrical insulator in high-voltage circuit breakers and transformers.

Compound	Hybridization	Geometry (VSEPR)	Hydrolysis & Reactivity
SF4	sp^3d (1 lone pair eq)	**See-Saw (Distorted tetrahedron)**	Hydrolyses violently: SF4 + 2 H_2O → **SO2 + 4 HF** (fluorinating agent)
SF6	$sp^{3d}2$ (0 lone pair)	**Regular Octahedral**	Completely inert to water and hot caustic alkali
SCl4	sp^3d (1 lone pair)	See-Saw	Thermally unstable; decomposes above -30°C: SCl4 → SCl2 + Cl_2
S_2Cl_2	sp^3 (bent per S)	Open-book / H_2O_2-like	Yellow liquid; reacts with water: 2 S_2Cl_2 + 2 H_2O → SO2 + 3 S + 4 HCl

Table 16.8 — Comparative structures and reactions of Sulphur halides.

2. Thionyl Chloride (SOCl2) & Sulphuryl Chloride (SO2Cl2)

- **Thionyl Chloride (SOCl2):** Pyramidal (sp^3, 1 lone pair, one S=O, two S-Cl). Darzens reaction (organic synthesis of alkyl chlorides): R-OH + SOCl2 --(pyridine)--→ **R-Cl + SO2↑ + HCl↑**. (Both by-products SO2 and HCl are gases, escaping spontaneously to leave pure alkyl chloride!).
- **Sulphuryl Chloride (SO2Cl2):** Tetrahedral (sp^3, two S=O, two S-Cl). Hydrolysis: SO2Cl2 + 2 H_2O → **H_2SO_4 + 2 HCl**.

Group 17: Halogens ($ns^2\ np^5$) — Salt Formers & Periodic Anomalies

Contains F, Cl (gases), Br (volatile red liquid), I (lustrous solid), At, Ts (radioactive). Highest effective nuclear charge in their respective periods, having the most negative electron gain enthalpies.

1. Periodic Trends of Halogens

Element	Config.	Covalent Radius (pm)	IE-1 (kJ/mol)	ΔH_{eg} (kJ/mol)	Bond Enthalpy X_2 (kJ/mol)
F	[He] $2s^2\ 2p^5$	64	1680	-328	**158.8 (Low!)**
Cl	[Ne] $3s^2\ 3p^5$	99	1256	**-349 (Most negative!)**	**242.6 (Highest!)**
Br	[Ar] $3d^{10}\ 4s^2\ 4p^5$	114	1142	-325	192.8
I	[Kr] $4d^{10}\ 5s^2\ 5p^5$	133	1008	-295	151.1

Table 17.1 — Physical and atomic properties of Group 17 elements.

KEY RESULT

Bond Dissociation Enthalpy Anomaly (Classic JEE Question): **Cl_2 (242.6) > Br_2 (192.8) > F_2 (158.8) > I_2 (151.1 kJ/mol)**. F_2 has a lower bond energy than both Cl_2 and Br_2 because of severe inter-electronic repulsions between the non-bonding electron pairs in the extremely small F_2 molecule (F-F distance = 143 pm)!

2. Oxidising Power & Electrode Potentials

Standard reduction potential: F_2 (+2.87 V) > Cl_2 (+1.36 V) > Br_2 (+1.09 V) > I_2 (+0.54 V). Fluorine is the most powerful oxidising agent in chemistry. Why F_2 > Cl_2 despite lower electron gain enthalpy? Standard reduction potential depends on the Born-Haber cycle parameters:

- 1. **Low Bond Dissociation Enthalpy:** $\Delta H_{diss}(F_2)$ is very low (+158.8 kJ/mol) → easy atomisation.
- 2. **Very High Hydration Enthalpy:** F^- is small with immense charge density, releasing huge hydration enthalpy: $\Delta H_{hyd}(F^-)$ = **-515 kJ/mol** (vs Cl^- = -381 kJ/mol).
- 3. **Conclusion:** High hydration energy and low bond dissociation energy more than compensate for the slightly less negative electron gain enthalpy of F_2!

3. Anomalous Behaviour of Fluorine

Only oxidation state is **-1** (no d-orbitals, most electronegative element). Maximum covalency is strictly 1 (or 2 in rare bridging/coordination). HF is a weak acid with high boiling point (293 K) due to zig-zag hydrogen-bonded chains (H-F...H-F...).

Extraction of Halogens — Moissan Electrolysis, Deacon Process & Br_2/I_2 Mining

Because F_2 is the strongest oxidant, no chemical reagent can oxidise F^- to F_2; it is produced exclusively by electrolysis.

1. Extraction of Fluorine (Dennis-Moissan Process)

- **Electrolyte:** Anhydrous molten mixture of **KF + 2 HF (KF·2HF)** at ~100°C in a Monel metal or steel vessel.
- **Why Anhydrous HF alone cannot be used:** Pure anhydrous liquid HF is a non-electrolyte (does not conduct electricity). Adding KF supplies K^+ and $[HF2]^-$ ions, making it conductive.
- **Electrodes:** Carbon (graphite) anode, steel vessel as cathode. A copper diaphragm separates anode and cathode to prevent explosive recombination of H_2 and F_2: · At Cathode: $2\ H^+ + 2e^- \rightarrow \mathbf{H_2\uparrow}$ · At Anode: $2\ F^- \rightarrow \mathbf{F_2\uparrow} + 2e^-$

2. Manufacture of Chlorine (Cl2)

Process	Reagents / Mechanism	Key Reaction Equation
Deacon's Process	Oxidation of HCl gas by atmospheric O_2 with CuCl2 catalyst (723 K)	$4\ HCl + O_2$ --(CuCl2, 723 K)--→ **$2\ Cl_2 + 2\ H_2O$**
Chlor-Alkali (Nelson / Castner-Kellner)	Electrolysis of aqueous brine (NaCl)	$2\ NaCl + 2\ H_2O$ --(electrolysis)--→ **Cl_2 (anode) + H_2 (cathode) + 2 NaOH**
Laboratory Prep	Oxidation of conc. HCl with MnO2 or KMnO4	$MnO2 + 4\ HCl \rightarrow$ **$MnCl2 + Cl_2\uparrow + 2\ H_2O$** 2 $KMnO4 + 16\ HCl \rightarrow$ **$2\ KCl + 2\ MnCl2 + 5\ Cl_2\uparrow + 8\ H_2O$**

Table 17.2 — Synthetic and industrial processes for Chlorine.

3. Recovery of Bromine and Iodine

- **Bromine from Sea Brine:** Sea water contains Br^- (65 ppm). Acidified with H^2SO^4, treated with Cl^2 gas: $^{**}2\ Br^{\wedge}-^{\wedge} + Cl\ 2 \rightarrow 2\ Cl^{\wedge}-^{\wedge} + Br^2\uparrow^{**}$. Blown out with air, absorbed in Na^2CO^3 solution: $3\ Na^2CO^3 + 3\ Br^2 \sim \rightarrow 5\ NaBr + NaBrO3 + 3\ CO2$.
- **Iodine from Caliche (Chile Saltpetre):** Contains sodium iodate NaIO3. Reduced with sodium bisulphite: **$2\ NaIO3 + 5\ NaHSO3 \rightarrow 2\ Na_2SO_4 + 3\ NaHSO4 + I_2\downarrow + H_2O$**.
- **Iodine from Seaweeds (Laminaria):** Ash (Kelp) contains iodides; boiled with conc. H_2SO_4 and MnO2: $2\ NaI + MnO2 + 3\ H_2SO_4 \rightarrow 2\ NaHSO4 + MnSO4 + \mathbf{I_2\uparrow} + 2\ H_2O$.

Hydrogen Halides (HX) — Boiling Points, Bond Energies, Acidity & Reducing Trends

Prepared by direct combination or by salt + acid reactions: NaX + H_2SO_4 → NaHSO4 + HX.

1. Periodic Trends in Hydrogen Halides

Property	HF	HCl	HBr	HI	Trend / Underlying Reason
Boiling Point (K)	**293 (Highest!)**	189	206	238	HF is anomalously high due to strong intermolecular H-bonding; then increases with van der Waals dispersion forces: **HCl < HBr < HI < HF**
Bond Length (pm)	91.7	127.4	141.4	**160.9**	Increases down group with increasing halogen size
Bond Enthalpy (kJ/mol)	**574**	432	363	295	Decreases sharply as overlap efficiency declines
Thermal Stability	**Most Stable**	Stable	Decomposes >800 K	**Decomposes readily**	Parallels bond dissociation energy: **HF > HCl > HBr > HI**
Acidic Strength (K_a)	Weak (3.5×10^-4)	Strong (~10^7)	Very Strong (~10^9)	**Strongest (~10^10)**	Follows inverse of H-X bond strength: **HF < HCl < HBr < HI**
Reducing Character	Nil	Very Weak	Strong	**Strongest**	HI easily oxidises to I_2: **HF < HCl < HBr < HI**

Table 17.3 — Master summary of Hydrogen Halide properties.

KEY RESULT

Critical Sequences for JEE: · **Boiling Point:** HCl HCl > HBr > HI

2. Preparation Traps with H2SO4

TRAP

TRAP: HBr and HI CANNOT be prepared by heating NaBr/NaI with conc. H_2SO_4! Conc. H_2SO_4 is a strong oxidising agent that oxidises HBr to Br_2 and HI to I_2: 2 NaBr + 2 H_2SO_4 → Na_2SO_4 + **Br_2 (red vapor)** + SO2 + 2 H_2O 2 NaI + 2 H_2SO_4 → Na_2SO_4 + **I_2 (violet vapor)** + SO2 + 2 H_2O. **Correct Method:** Use non-oxidising **conc. H_3PO_4**: 3 NaBr + H_3PO_4 --(Δ)--→ Na_3PO_4 + 3 HBr↑.

Halogen Oxides — OF2, Paramagnetic ClO2, Dimeric Cl_2O_6 & I_2O_5 in CO Estimation

Fluorine forms oxygen fluorides (F is more electronegative than O). Other halogens form oxides.

1. Oxygen Fluorides (OF2 and O2F2)

- **OF2 (Oxygen Difluoride):** Prepared: 2 NaOH (cold, 2% dil.) + 2 F_2 → **OF2**↑ + 2 NaF + H_2O. V-shaped (angle 103°, F-O-F repulsions compress angle below water's 104.5°). Oxidation state of O is **+2**.
- **O_2F_2 (Dioxygen Difluoride):** Prepared: F_2 + O_2 in electric discharge at 90 K. Open-book structure like H $_2O_2$. Oxidation state of O is **+1**. Extremely powerful fluorinating agent; reacts with plutonium: **Pu + 3 O_2 F_2 → PuF6 + 3 O_2** (used to recover Pu from spent nuclear fuel!).

2. Oxides of Chlorine

Oxide	Oxid. State	Structure & Geometry	Key Reaction / Diagnostic Behavior
Cl_2O	+1	Bent V-shaped (angle 111°)	2 HgO + 2 Cl_2 → HgO·HgCl2 + **Cl_2O**; Anhydride of HOCl
ClO2	+4	Bent (odd-electron, 19 e^-)	**Paramagnetic gas**; does not dimerise readily; bleaching agent for wood pulp; 2 ClO2 + 2 NaOH → **NaClO2 + NaClO3** + H_2O (mixed anhydride!)
Cl_2O_6	+6	**Liquid / Solid: $[ClO_2]^+$ $[ClO_4]^-$**	Mixed anhydride of HClO3 and HClO4: Cl_2O_6 + H_2O → **HClO3 + HClO4**
Cl_2O_7	+7	Two ClO3 tetrahedra sharing O bridge	Anhydride of perchloric acid: 2 HClO4 + P_4O_{10} → **Cl_2O_7** + 4 HPO3. Colorless oily liquid, explodes violently on percussion!

Table 17.4 — Comprehensive spectrum of Chlorine Oxides.

3. Iodine Pentoxide (I2O5) & Quantitative Estimation of CO

KEY RESULT

Prepared by dehydrating iodic acid: 2 HIO3 --(440 K)--→ **I_2O_5** + H_2O. White crystalline solid, highly stable. Reacts quantitatively with Carbon Monoxide at 170°C: **I_2O_5 + 5 CO → I_2 + 5 CO2**. The liberated I $_2$ is titrated against standard sodium thiosulphate ($Na_2S_2O_3$) with starch indicator. This is the standard analytical method for estimating toxic CO in air and flue gases!

Oxoacids of Halogens — Acid Strengths, Anion Resonance & Thermal Stability

Halogens form four series of oxoacids: Hypohalous (HOX, +1), Halous (HXO2, +3), Halic (HXO3, +5), Perhalic (HXO4, +7).

1. Complete Classification of Halogen Oxoacids

Oxidation State	Acid Family	Fluorine	Chlorine	Bromine	Iodine
+1	Hypohalous (HOX)	HOF (only one!)	HOCl	HOBr	HOI
+3	Halous (HXO2)	—	HClO2	—	—
+5	Halic (HXO3)	—	HClO3	HBrO3	HIO3
+7	Perhalic (HXO4)	—	HClO4	HBrO4	HIO4 / H_5IO_6 (Periodic)

Table 17.5 — Systematic matrix of halogen oxoacids.

2. Acid Strength Trends

- **Trend 1 (Same Halogen, Increasing Oxidation State): HOCl (+1) < HClO2 (+3) < HClO3 (+5) < HClO4 (+7)**. Reason: Conjugate base stability increases as the negative charge is delocalised over more electronegative oxygen atoms via resonance: ClO^- (1 O) < ClO_2^- (2 O) < ClO_3^- (3 O) < **ClO_4^- (4 O, charge density −0.25/O)**. HClO4 is one of the strongest known Bronsted acids!
- **Trend 2 (Different Halogens, Same Oxidation State): HOCl > HOBr > HOI**. Reason: As electronegativity of halogen decreases (Cl > Br > I), polarity of the O-H bond decreases, making release of H^+ less favorable.

3. Oxidising Power vs Thermal Stability

Family Sequence	Acid Strength Order	Thermal Stability Order	Oxidising Power Order
Chlorine Series	HOCl < HClO2 < HClO3 < **HClO4**	HOCl < HClO2 < HClO3 < **HClO4**	**HOCl** > HClO2 > HClO3 > HClO4

Table 17.6 — Inverse relationship between stability and oxidising power.

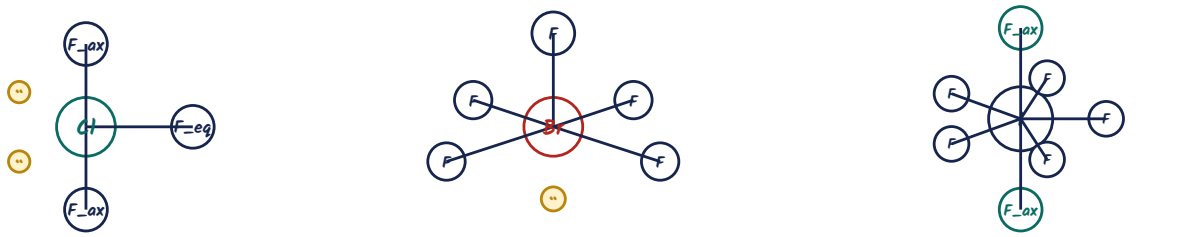
KEY RESULT

Oxidising Power is INVERSE to acid stability: Hypochlorous acid **HOCl is the STRONGEST oxidant**, whereas perchloric acid **HClO4 is the weakest oxidant** in dilute aqueous solution!

Interhalogen Compounds (XX'_n) — Synthesis, VSEPR Geometries & Reactivity

Formed when two different halogens react. General formula XX'_n where X is the larger halogen and X' is the smaller, more electronegative halogen; n = 1, 3, 5, 7.

1. Comprehensive VSEPR Geometries of Interhalogens



ClF_3 (T-Shaped, sp^3d)
2 lp equatorial (87.5° bend)

BrF_5 (Sq. Pyramidal, sp^3d^2)
1 lp axial; F-Br-F angle 84.8°

IF_7 (Pentagonal Bipyramidal, sp^3d^3)
Axial (90°): 179 pm; Eq (72°): 186 pm

Fig. 17.1 — Interhalogen geometries: ClF3 (T-Shaped), BrF5 (Square Pyramidal), and IF7 (Pentagonal Bipyramidal).

Type	Example	Hybridization	Lone Pairs on X	VSEPR Geometry	Physical State
XX'	ClF, BrF, ICl	sp^3	3 lone pairs	**Linear**	ClF (gas), ICl (ruby red solid)
XX'_3	ClF3, BrF3, ICl3	sp^3d	2 equatorial lp	**T-Shaped** (bent T)	ClF3 (gas), BrF3 (liquid), ICl3 (dimer solid)
XX'_5	ClF5, BrF5, IF5	sp^3d^2	1 axial lp	**Square Pyramidal**	All colorless liquids/gases
XX'_7	**IF7** (only one!)	sp^3d^3	0 lone pairs	**Pentagonal Bipyramidal**	Colorless gas (D_{5h} symmetry)

Table 17.7 — Complete structural geometries of interhalogens.

KEY RESULT
Why is IF7 the ONLY XX'_7 compound formed? Because Iodine has the largest atomic radius in Group 17, and Fluorine has the smallest atomic radius. The radius ratio r(I)/r(F) is maximum, allowing sterically uncrowded coordination of seven F atoms!

2. Chemical Reactivity: Interhalogens vs Halogens

In general, **interhalogen compounds are MORE reactive than pure halogens (except F_2)**. Reason: The X-X' bond in an interhalogen is polar and weaker than the non-polar X-X bond in a pure halogen (except F-F where lone pair repulsions weaken the bond).

3. Hydrolysis of Interhalogens

Hydrolysis always yields a **halide ion from the smaller halogen X'** and an **oxohalate ion from the larger central halogen X**: · **$XX' + H_2O \rightarrow HX' + HOX$** (e.g., $ICl + H_2O \rightarrow HCl + HOI$) · **$XX'_3 + 2\ H_2O \rightarrow 3\ HX' + HXO2$** (e.g., $BrF3 + 2\ H_2O \rightarrow 3\ HF + HBrO2$) · **$XX'_5 + 3\ H_2O \rightarrow 5\ HX' + HXO3$** (e.g., $IF5 + 3\ H_2O \rightarrow 5\ HF + HIO3$) · **$XX'_7 + 4\ H_2O \rightarrow 7\ HX' + HXO4$** (e.g., $IF7 + 4\ H_2O \rightarrow 7\ HF + H_5IO_6$)

TRAP
TRAP: ICl3 in the solid state exists as a planar bridged dimer I_2Cl_6 with two bridging chlorine atoms (chlorine-bridged like Al_2Cl_6)!

Polyhalide Ions — Triiodide I_3^-, ICl_2^-, ICl_4^- & 3c-4e Hypervalent Bonding

Halogens and interhalogens coordinate with halide ions to form polyhalide anions: $X^- + n\,X_2 \rightarrow X_{2n+1}^-$.

1. Triiodide Ion (I3-) — Structure & Bonding

- **Solubility of I_2 in KI:** Pure iodine is virtually insoluble in pure water (non-polar). It dissolves readily in aqueous KI due to formation of the soluble **triiodide complex anion**: $KI + I_2 \rightleftharpoons K^+[I_3]^-$.
- **VSEPR Geometry:** Central I atom has 7 valence e^- + 1 from charge + 2 from bond pairs = 10 electrons = 5 pairs (sp^3d hybridization).
- **Three Equatorial Lone Pairs:** The three lone pairs occupy equatorial positions at 120° to minimise repulsions. The two I-I bonds are axial (180°). Geometry is **perfectly LINEAR ($D_{\infty h}$ symmetry)** with I-I bond length 292 pm.

Polyhalide Ion	Central Atom	Hybridization	Lone Pairs	Geometry (VSEPR)	Bond Angle
I_3^-	I	sp^3d	3 equatorial	**Linear**	180°
ICl_2^-	I	sp^3d	3 equatorial	**Linear**	180°
ICl_4^-	I	sp^3d^2	2 axial	**Square Planar**	90°
I_3^+ (Cation)	I	sp^3	2 lone pairs	**Bent / V-shaped**	104°
I_5^-	I	—	—	**V-shaped dimer** (I_3^- coordinated to I_2)	~95°

Table 17.8 — Geometries and electron pairs of Polyhalide species.

KEY RESULT
Why does Fluorine NOT form polyhalide anions (F_3^- does not exist)? Because Fluorine has no vacant d orbitals to expand its octet to accommodate 10 valence electrons (sp^3d)!

2. Polyhalide Cations

Strong oxidising Lewis acids like AsF5 or SbF5 can pull F^- from interhalogens to generate polyhalide cations: · $ClF_3 + SbF_5 \rightarrow$ **$[ClF_2]^+[SbF_6]^-$** (bent $[ClF_2]^+$, sp^3, 2 lone pairs) · $BrF_3 + SbF_5 \rightarrow$ $[BrF_4]^+[SbF_6]^-$ (see-saw $[BrF_4]^+$, sp^3d, 1 lone pair)

Pseudohalogens & Pseudohalides — Cyanogen (CN)2, Thiocyanogen (SCN)2 & Chemistry

Pseudohalogens are univalent polyatomic groupings that resemble true halogens (X_2) in their free state, forming univalent anions (pseudohalides X^-) and volatile hydrogen acids (HX).

1. Defining Criteria for Pseudohalogens

- 1. They consist of two or more electronegative atoms, at least one of which is Nitrogen.
- 2. They dimerise in the free state to neutral symmetric volatile dimers: **(CN)2 (Cyanogen)**, **(SCN)2 (Thiocyanogen)**, **(SeCN)2**, **(OCN)2**.
- 3. They react with alkalis to disproportionate into pseudohalide and pseudohalite (like $Cl_2 + 2\ OH^- \rightarrow Cl^- + ClO^- + H_2O$): **$(CN)_2 + 2\ OH^- \rightarrow CN^-$ (Cyanide) + OCN^- (Cyanate) + H_2O**.
- 4. They form insoluble silver, lead, and mercury(I) salts: AgCN (white↓), AgSCN (white↓), resembling AgCl↓.

Pseudohalide Ion	Formula	Dimeric Pseudohalogen	Insoluble Ag^+ Salt & Behaviour in NH3
Cyanide	CN^-	(CN)2 (Cyanogen, gas)	AgCN↓ (white, dissolves in excess NH3 and in excess CN^-)
Thiocyanate	SCN^-	(SCN)2 (Thiocyanogen, yellow oil)	AgSCN↓ (white, insoluble in dil. HNO3; reacts with Fe^{3+} → blood-red $[Fe(SCN)]^{2+}$)
Cyanate	OCN^-	(OCN)2	AgOCN↓ (white)
Azide	N_3^-	$(N_3)_2$ (Does NOT exist!)	AgN3↓ (white, shock-sensitive explosive detonator)

Table 17.9 — Classification of Pseudohalogens and Pseudohalides.

KEY RESULT

TRAP Question: Is Azide (N_3^-) a true pseudohalide? N_3^- is an **incomplete pseudohalide** because the free neutral dimer '$(N_3)_2$' (hexanitrogen N_6) does NOT exist as a stable isolable molecule!

Bleaching Powder & Disproportionation — CaOCl2, Available Chlorine & Alkali Reactions

Bleaching powder is manufactured by action of chlorine on dry slaked lime: $Ca(OH)_2 + Cl_2 \rightarrow CaOCl2{\cdot}H_2O$.

1. Composition & Chemical Nature of Bleaching Powder

- **True Composition:** Bleaching powder is NOT a simple single compound CaOCl2. Its actual structural formula is a mixed double salt: $\mathbf{Ca(OCl)_2{\cdot}CaCl_2{\cdot}Ca(OH)_2{\cdot}2H_2O}$.
- **Oxidation State of Chlorine in CaOCl2:** CaOCl2 dissociates into Ca^{2+}, Cl^-, and OCl^- ions. · Direct Chloride ion (Cl^-): **Oxidation state = −1** · Hypochlorite ion (OCl^-): **Oxidation state = +1** · Average oxidation state = **0**! (Classic JEE question).
- **Auto-Decomposition on Long Standing:** 6 CaOCl2 --(standing in air)--→ $\mathbf{Ca(ClO_3)_2 + 5\ CaCl2}$ (loss of bleaching power).

2. Available Chlorine & Iodometric Titration

The percentage of active chlorine liberated by treatment with excess dilute acid is called **Available Chlorine**: CaOCl2 + 2 CH3COOH (or dil. H_2SO_4) → $Ca(CH_3COO)_2 + H_2O + \mathbf{Cl_2}\uparrow$. The liberated Cl_2 is reacted with excess KI: $Cl_2 + 2\ KI \rightarrow 2\ KCl + \mathbf{I_2}$. The liberated I_2 is titrated against standard 0.1 N $Na_2S_2O_3$ with starch indicator: $\mathbf{I_2 + 2\ Na_2S_2O_3 \rightarrow 2\ NaI + Na_2S_4O_6}$.

3. Master Summary of Halogen Disproportionation with Alkali

Halogen	With Cold, Dilute NaOH (15°C)	With Hot, Concentrated NaOH (70°C)
Cl_2	$Cl_2 + 2\ NaOH \rightarrow$ **NaCl (−1) + NaOCl (+1)** + H_2O	$3\ Cl_2 + 6\ NaOH \rightarrow$ **5 NaCl (−1) + NaClO3 (+5)** + 3 H_2O
Br_2	$Br_2 + 2\ NaOH \rightarrow$ **NaBr + NaOBr** + H_2O (unstable)	$3\ Br_2 + 6\ NaOH \rightarrow$ **5 NaBr (−1) + NaBrO3 (+5)** + 3 H_2O
I_2	Disproportionates instantly to iodate even in cold!	$3\ I_2 + 6\ NaOH \rightarrow$ **5 NaI (−1) + NaIO3 (+5)** + 3 H_2O
F_2 (Redox!)	**$2\ F_2 + 2\ NaOH$ (dil) → 2 NaF + OF2↑ + H_2O**	**$2\ F_2 + 4\ NaOH$ (conc) → 4 NaF + O_2↑ + 2 H_2O**

Table 17.10 — Reaction of Halogens with Cold/Dilute vs Hot/Concentrated Alkali.

KEY RESULT

Fluorine does NOT disproportionate with alkali! Because F cannot have positive oxidation states, it oxidises hydroxide oxygen: **cold dil. NaOH gives OF2**, and **hot conc. NaOH gives O_2**!

Group 18: Noble Gases ($ns^2\ np^6$) — Periodic Trends, Liquid Helium & Clathrates

Contains He, Ne, Ar, Kr, Xe, Rn (radioactive). Completely filled valence shells confer extraordinary chemical inertness, extremely high ionization energies, and zero electron affinity.

1. Periodic Trends in Group 18

Element	Config.	Atomic Radius (pm)	IE-1 (kJ/mol)	ΔH_{eg} (kJ/mol)	Boiling Point (K)
He	1s^2	99	**2372 (Highest in PT!)**	+48	**4.2 (Lowest in PT!)**
Ne	[He] $2s^2\ 2p6$	160	2080	+116	27.1
Ar	[Ne] $3s^2\ 3p6$	190	1520	+96	87.2
Kr	[Ar] $3d^{10}\ 4s2$ 4p^6	200	1351	+96	119.7
Xe	[Kr] $4d^{10}\ 5s2$ Sp^6	220	1170	+77	165.0
Rn	[Xe] $4f^{14}\ 5d10\ 6s^2\ 6p6$	230	1037	+68	211.0

Table 18.1 — Physical and atomic properties of Noble Gases.

KEY RESULT
Helium Records in Chemistry: · **Highest Ionization Enthalpy** in the entire periodic table (2372 kJ/mol). · **Lowest Boiling Point** of any known substance (4.2 K at 1 atm). · Does not solidify even at absolute zero (0 K) under atmospheric pressure; requires ~25 atm pressure!

2. Liquid Helium & Superfluidity (Helium II)

Liquid helium undergoes a second-order phase transition at **2.18 K (the λ-point)** from normal liquid Helium I to **Helium II (Superfluid)**: · **Zero Viscosity:** Flows through sub-microscopic capillaries without friction; creeps up and over container walls in a thin Rollin film. · **Enormous Thermal Conductivity:** ~800 times higher than copper! Liquid Helium II does not boil with bubbles; evaporation occurs silently from the surface.

3. Clathrate (Cage / Inclusion) Compounds

- **Definition:** Non-stoichiometric guest-host inclusion compounds where noble gas atoms (guest) are physically trapped inside the hydrogen-bonded cage cavities of host molecules (quinol / hydroquinone or ice), without any chemical bonds.
- **Conditions for Clathrate Formation:** · Host: Quinol (p-dihydroxybenzene) crystallised under pressure of noble gas. · **Ar, Kr, Xe form clathrates** because their atomic radii (1.9 to 2.2 Å) fit snugly into the 4.0 Å cavities. · **He and Ne do NOT form clathrates!** He and Ne atoms are too small and easily slip through the cavity walls without being trapped.

Xenon Fluorides (XeFn) — Bartlett's Discovery, VSEPR Geometries & Hydrolysis

In 1962, Neil Bartlett noticed that O_2 reacts with PtF6 to form $[O_2]^+$ $[PtF6]^-$. Knowing the first IE of Xe (1170 kJ/mol) is almost identical to that of O_2 (1175 kJ/mol), he reacted Xe with PtF6 to synthesize the first noble gas compound: **Xe^+ $[PtF6]^-$ (orange-yellow solid)**.

1. Controlled Direct Synthesis of Xenon Fluorides

Compound	Xe : F_2 Ratio	Conditions (T, P)	Physical State & Geometry
XeF2	**2 : 1 (Excess Xe)**	673 K, 1 bar (Nickel tube)	White solid, sublimes at 298 K; **Linear** (sp^3d)
XeF4	**1 : 5 Ratio**	873 K, 7 bar (Nickel tube)	White crystalline solid; **Square Planar** (sp^3d^2)
XeF6	**1 : 20 (Excess F_2)**	573 K, 60-70 bar	Colorless crystals (yellow liquid); **Distorted Octahedral**

Table 18.2 — Synthetic parameters for Xenon Fluorides (NCERT core).

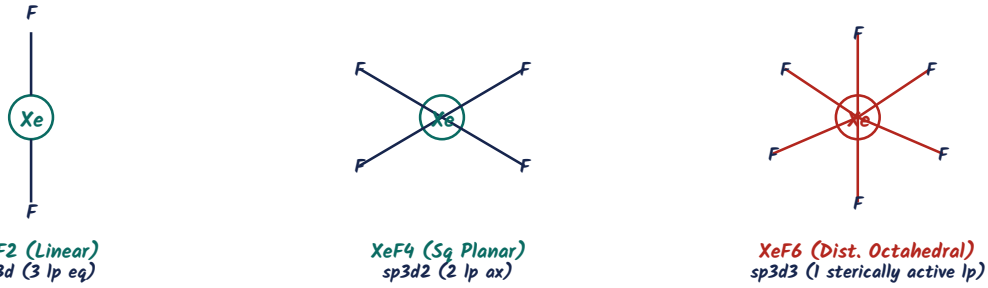


Fig. 18.1 — VSEPR geometries of XeF2 (Linear), XeF4 (Square Planar), XeF6 (Distorted Octahedral).

2. Hydrolysis Pathways of Xenon Fluorides (Crucial JEE Topic)

- **1. Hydrolysis of XeF2 (Redox): $2\ XeF_2 + 2\ H_2O \rightarrow 2\ Xe\uparrow + 4\ HF + O_2\uparrow$** (Xe is reduced +2 → 0; O is oxidised -2 → 0).
- **2. Hydrolysis of XeF4 (Disproportionation): $6\ XeF_4 + 12\ H_2O \rightarrow 2\ XeO_3$ (+6) + 4 Xe (0) + 24 HF + 3 $O_2\uparrow$** (Xe(+4) disproportionates into Xe(+6) in XeO3 and Xe(0) gas!).
- **3. Hydrolysis of XeF6 (Non-Redox, Stepwise):** · Partial hydrolysis (1 eq H_2O): **$XeF_6 + H_2O \rightarrow XeOF_4$ (colorless liquid) + 2 HF** · Partial hydrolysis (2 eq H_2O): **$XeF_6 + 2\ H_2O \rightarrow XeO_2F_2$ (colorless solid) + 4 HF** · Complete hydrolysis (3 eq H_2O): **$XeF_6 + 3\ H_2O \rightarrow XeO_3$ (explosive solid) + 6 HF** (All retain Xe in +6 state!).

TRAP
TRAP: Hydrolysis of XeF4 is a DISPROPORTIONATION reaction that evolves Xe gas, whereas complete hydrolysis of XeF6 is NON-REDOX and produces explosive solid XeO3 without liberating Xe gas!

Xenon Oxides & Oxofluorides — VSEPR Geometries, Hybridization & Explosive Nature

Xenon forms explosive oxides XeO3, XeO4, and intermediate oxofluorides XeOF4, XeO2F2, XeOF2.

1. Comprehensive Structural Matrix of Xenon-Oxygen Compounds

Compound	Xe Oxid. State	Hybridization	Lone Pairs	VSEPR Shape	Bond Parameters & Angles
XeO3	+6	sp^3	1 lone pair	**Pyramidal** (trigonal)	Three Xe=O bonds (dπ-pπ), angle 103°
XeO4	+8	sp^3	0 lone pairs	**Regular Tetrahedral**	Four Xe=O double bonds, angle 109.5°
XeOF4	+6	sp^3d^2	1 lone pair (trans to =O)	**Square Pyramidal**	Four Xe-F basal + one Xe=O apical
XeO2F2	+6	sp^3d	1 equatorial lp	**See-Saw**	Two equatorial Xe=O + two axial Xe-F
XeO3F2	+8	sp^3d	0 lone pairs	**Trigonal Bipyramidal**	Three equatorial Xe=O + two axial Xe-F

Table 18.3 — Complete VSEPR geometries of Xenon Oxides and Oxofluorides.

KEY RESULT

Pi Bonding in Xenon Oxides: All Xe=O double bonds in xenon chemistry are formed by **dπ-pπ overlap** (donation of oxygen 2p electrons into empty xenon 5d orbitals). For example, XeO3 has 3 dπ-pπ bonds, and XeO4 has 4 dπ-pπ bonds!

2. Xenon Trioxide (XeO3) & Xenates

- **Properties of XeO3:** Colorless, hygroscopic, highly explosive crystalline solid. Detonates on touch or heating: $2\ XeO_3(s) \xrightarrow{(shock)} 2\ Xe(g) + 3\ O_2(g)$ ($\Delta H° = -402$ kJ/mol, violent explosion!).
- **Reaction with Base (Xenate Ion):** XeO_3 dissolves in aqueous base to form xenate ion: $XeO_3 + OH^- \rightleftharpoons$ **$[HXeO_4]^-$ (Hydrogen xenate ion)**. In basic solution, xenate disproportionates slowly into **Perxenate $[XeO_6]^{4-}$** and Xenon gas: $2\ [HXeO_4]^- + 2\ OH^- \rightarrow$ **$[XeO_6]^{4-}$ (Perxenate, +8 state)** + **$Xe + O_2 + 2\ H_2O$**.
- **Barium Perxenate Ba_2XeO_6:** Stable salt. Reacts with cold conc. H_2SO_4 to yield volatile, highly explosive gaseous **XeO_4 (Xenon tetroxide)**: $Ba_2XeO_6 + 2\ H_2SO_4 \rightarrow 2\ BaSO_4\downarrow +$ **$XeO_4\uparrow$** $+ 2\ H_2O$.

Xenon Fluoride Complexes — Fluoride Ion Acceptors vs Donors

Xenon fluorides react with fluoride ion acceptors (Lewis acids) to form fluoro-cations, and with fluoride ion donors (alkali metal fluorides) to form fluoro-anions.

1. Fluoride Ion Donor Reactions (Forming Xenon Cations)

With strong covalent Lewis acids (MFS where M = P, As, Sb, Bi, Ru, Pt, Nb, Ta), XeFn acts as a **fluoride ion donor** [F^- donor]:

- · $XeF_2 + PFS \rightarrow$ **$[XeF]^+ [PF6]^-$** ($[XeF]^+$ is linear)
- · $XeF_2 + 2\ SbFS \rightarrow$ **$[XeF]^+ [Sb_2F_{11}]^-$**
- · $XeF_4 + SbFS \rightarrow$ **$[XeF3]^+ [SbF6]^-$** ($[XeF3]^+$ is T-shaped, sp^3d with 2 lone pairs)
- · $XeF_6 + AsFS \rightarrow$ **$[XeFS]^+ [AsF6]^-$** ($[XeFS]^+$ is square pyramidal, $sp^{3d}2$ with 1 lone pair)

Xenon Fluoride	Lewis Acid / Base	Product Salt Formed	Role of Xenon Fluoride
XeF_2	PFS / SbFS	$[XeF]^+ [PF6]^-$ / $[XeF]^+ [SbF6]^-$	**F^- Donor (Lewis Base)**
XeF_4	SbFS	$[XeF3]^+ [SbF6]^-$	**F^- Donor (Lewis Base)**
XeF_6	AsFS / SbFS	$[XeFS]^+ [AsF6]^-$ / $[XeFS]^+ [SbF6]^-$	**F^- Donor (Lewis Base)**
XeF_6	CsF / RbF (alkali fluorides)	$Cs^+ [XeF7]^-$ / $Cs_2[XeF8]$	**F^- Acceptor (Lewis Acid!)**

Table 18.4 — Dual Lewis acid-base behaviour of Xenon Fluorides.

KEY RESULT
Dual Nature of XeF6: XeF6 is amphoteric in the Lewis sense: it can donate F^- to SbFS to form **$[XeFS]^+$**, and it can accept F^- from alkali fluorides MF to form **$[XeF7]^-$** or **$[XeF8]^{2-}$** (Octafluoroxenate(VI) ion)!

2. Geometry of [XeF8]2- Anion

In $Cs_2[XeF8]$ and NO2[XeF8], the $[XeF8]^{2-}$ anion has 8 bond pairs and 1 lone pair. Its structural geometry is a **Square Antiprismatic** arrangement (D_{4d} symmetry). It is thermally stable up to 400°C.

Krypton, Radon & JEE Applications — KrF2, Radon Chemistry & Industrial Uses

Beyond Xenon, chemical reactivity in Group 18 is severely limited by ionization energy.

1. Chemistry of Krypton & Radon

- **Krypton Difluoride (KrF2):** The only well-characterised krypton compound. Prepared by electrical discharge or UV irradiation of Kr + F_2 at -196°C: Kr + F_2 --(glow discharge, 77 K)--→ **KrF2 (colorless volatile solid)**. Linear structure (sp^{3d}, 3 equatorial lone pairs). Thermally unstable (decomposes at 0°C). It is an even **stronger fluorinating agent than XeF2**: Ag + KrF2 → AgF2; Au + KrF2 + SbFS → **[AuF6]_^ **(oxidises Gold to Au(+V)!)**.
- **Radon Chemistry:** Radon is radioactive (Rn-222, t_1/2 = 3.82 days, α-emitter). Forms **RnF2** (identified by radiochemical tracer and electromigration techniques). Covalent chemistry is difficult to study due to intense radiolysis.
- **Helium, Neon, Argon:** No neutral isolable chemical compounds known at standard conditions. Molecular ions like HeH^+, He_2^+, ArH^+ detected only in mass spectrometers.

2. Master Summary of Noble Gas Uses in JEE

Gas	Physical Property Utilised	Industrial, Laboratory & Medical Applications
Helium (He)	Low density & non-inflammable	Filling weather balloons and airships (safe substitute for H_2)
Helium (He)	Low solubility in blood at high pressure	Deep-sea diving gas cylinders (mixture of **80% He + 20% O_2**) to prevent the painful decompression sickness ('the bends')
Helium (He)	Extremely low boiling point (4.2 K)	Cryogenic coolant for superconducting magnets in **MRI (Magnetic Resonance Imaging)** and NMR
Neon (Ne)	Bright orange-red glow in discharge tubes	**Neon advertising display signs** and beacon lights for airfield navigation (penetrates fog)
Argon (Ar)	Complete chemical inertness & cheap	Inert gas atmosphere in metallurgical arc-welding of Al and stainless steel; filling incandescent light bulbs
Xenon & Krypton	Intense bright flash on discharge	High-speed photography strobe flashbulbs; lasers
Radon (Rn)	Radioactive α-emission	Radiotherapy for cancer treatment; hydrological tracing

Table 18.5 — Practical applications of Noble Gases in JEE.

Master Cross-Group Review — Comparative Periodic Trends Across Groups 13 to 18

A consolidated comparison of anomalous trends, inert pair redox patterns, and Lewis acidity across all six p-block groups.

1. Atomic Radii & Ionization Energy Anomalies Summary

Property / Trend	Standard Group Expectation	Actual Observed Sequence & Underlying Physical Reason
Atomic Radius in G13	B < Al < Ga < In < Tl	**B < Ga < Al < In ≈ Tl** (Ga < Al due to 3d transition contraction, d^10 poor shielding)
Ionization Enthalpy in G13	B > Al > Ga > In > Tl	**B > Tl > Ga > Al > In** (Tl > In due to 4f lanthanoid contraction, high Z_eff)
Ionization Enthalpy in G14	C > Si > Ge > Sn > Pb	**C > Si > Ge > Pb > Sn** (Pb > Sn due to 4f contraction)
IE-1: G15 vs G16	Group 16 > Group 15	**Group 15 > Group 16** (N > O, P > S, As > Se due to stable half-filled np^3 configuration)
Electron Gain Enthalpy (G16)	O > S > Se > Te	**S > Se > Te > Po > O** (Oxygen is LEAST negative due to compact 2p inter-electronic repulsions)
Electron Gain Enthalpy (G17)	F > Cl > Br > I	**Cl > F > Br > I** (Chlorine is MOST negative in entire periodic table; F has compact 2p repulsions)
Bond Dissociation Energy (G17)	$F_2 > Cl_2 > Br_2 > I_2$	**$Cl_2 > Br_2 > F_2 > I_2$** (F_2 has low bond energy due to severe lone-pair repulsions, 143 pm)

Table R.1 — Master compendium of p-block periodic anomalies.

2. The Inert Pair Effect Matrix Across p-Block

Group	Lower State (+n)	Higher State (+n+2)	Oxidising Agent (+n+2 State)	Reducing Agent (+n State)
Group 13	+1 (Tl^{+} stable)	+3 (B^{3+}, Al^{3+} stable)	**Tl^{3+} is strongly oxidising** ($Tl^{3+} + 2e^{-} \rightarrow Tl^{+}$)	Al^{+}, Ga^{+} (disproportionate)
Group 14	+2 (Pb^{2+} stable)	+4 (C^{4+}, Si^{4+} stable)	**Pb^{4+} is strongly oxidising** ($PbO2 + 4\,HCl \rightarrow PbCl2 + Cl_2$)	**Sn^{2+} is strongly reducing** ($Sn^{2+} \rightarrow Sn^{4+} + 2e^{-}$)
Group 15	+3 (Bi^{3+} stable)	+5 (N^{5+}, P^{5+} stable)	**Bi^{5+} is strongly oxidising** ($NaBiO3$ oxidises Mn^{2+} to $MnO4^{-}$)	P^{3+}, As^{3+} (mild reducing)

Table R.2 — Comparative redox chemistry governed by the inert pair effect.

KEY RESULT

Inert Pair Generalisation: · The lower oxidation state becomes progressively **MORE stable** descending the group: Tl(+) > In(+) > Ga(+) > Al(+) > B(+). · The higher oxidation state becomes progressively **A STRONGER OXIDANT** descending the group: Tl(3+) > In(3+) > Ga(3+) > Al(3+).

Complete Hydrolysis Map — Mechanisms, Redox vs Non-Redox & Intermediate Acids

Hydrolysis in the p-block occurs via two distinct pathways: nucleophilic attack on empty d orbitals (SN2-type), or coordination to an empty p orbital (Lewis acid-base). Below is the comprehensive master reaction map.

1. Non-Redox Hydrolysis (Oxidation State Preserved)

Substrate	Hydrolysis Equation	Nature of Attack / Intermediate
BCl3, BBr3, BI3	$BX3 + 3\ H_2O \rightarrow$ **H_3BO_3 + 3 HX**	Water attacks vacant 2p orbital on Boron
BF3	**4 BF3 + 3 H_2O → H_3BO_3 + 3 H[BF4]**	HF formed coordinates with unreacted BF3 to give fluoroboric acid!
SiCl4	$SiCl4 + 4\ H_2O \rightarrow$ **$Si(OH)_4$ (Silicic acid) → SiO2·2H2O + 4 HCl**	Attack on vacant 3d orbitals; coordination number expands to 6
PCl3	$PCl3 + 3\ H_2O \rightarrow$ **H_3PO_3 (Phosphorous acid) + 3 HCl**	Preserves +3 state; lone pair remains on P
PCl5	$PCl5 + H_2O \rightarrow$ **POCl3** + 2 HCl; $POCl3 + 3\ H_2O \rightarrow$ **H_3PO_4 + 3 HCl**	Stepwise via phosphorus oxychloride
SF4	$SF4 + 2\ H_2O \rightarrow$ **SO2 + 4 HF**	Water attacks vacant d-orbitals; preserves +4 state
XeF6	$XeF6 + H_2O \rightarrow$ **XeOF4** + 2 HF → **XeO2F2** + 4 HF → **XeO3 + 6 HF**	Stepwise replacement of F_2 by =O; all preserve Xe(+6)

Table R.3 — Master non-redox hydrolysis reactions.

2. Redox & Disproportionation Hydrolysis Reactions

Substrate	Reaction Equation	Type of Redox Reaction
NCl3	$NCl3 + 3\ H_2O \rightarrow$ **NH3 (-3) + 3 HOCl (+1)**	Non-redox on paper, but N acts as negative end ($\chi_N > \chi_{Cl}$)! Cl goes to HOCl, NOT HCl!
P_4 (in alkali)	$P_4 + 3\ NaOH + 3\ H_2O \rightarrow$ **PH3 (-3) + 3 NaH2PO2 (+1)**	Disproportionation of White Phosphorus
Cl_2 (in water)	$Cl_2 + H_2O \rightleftharpoons$ **HCl (-1) + HOCl (+1)**	Disproportionation (forms chlorine water; turns blue litmus red then bleaches)
XeF2	**2 XeF2 + 2 H_2O → 2 Xe (0) + 4 HF + O_2 (0)**	**Redox:** Xe(+2) reduced to 0; H_2O(-2) oxidised to O_2(0)
XeF4	**6 XeF4 + 12 H_2O → 2 XeO3 (+6) + 4 Xe (0) + 24 HF + 3 O_2↑**	**Disproportionation + Redox:** Xe(+4) splits into +6 and 0

Table R.4 — Redox and disproportionation hydrolysis pathways in JEE.

KEY RESULT

The Classic NCl3 vs NF3 vs PCl3 Hydrolysis Trap: · **NF3:** Does NOT hydrolyse at all (no vacant d-orbitals, high bond strength, non-polar). · **NCl3:** Hydrolyses to **NH3 + HOCl** (because $\chi_N = 3.0 > \chi_{Cl} = 2.8$; water attacks Cl with empty 3d orbitals!). · **PCl3:** Hydrolyses to **H_3PO_3 + HCl** (because $\chi_P = 2.1 < \chi_{Cl} = 3.0$; water attacks P with empty 3d orbitals!).